

When Targeted Grounded Gains Hide Complementary Abstraction Regression: DION-Bench and a Conjunctive Evaluation Protocol

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Abstract

Post-training with internal feedback can improve a targeted knowledge-grounded behavior while silently degrading complementary abstraction and logical-consistency capabilities. We propose a conjunctive evaluation protocol that accepts a method only when a designated target improves *and* complementary probes remain non-inferior (with a stricter secondary criterion requiring complementary equivalence), while also reporting retention and predictive drift. We instantiate the protocol in **DION-Bench**, combining two lexical/ontological probes (same-category exceptions; abstraction recovery) with a synthetic SAT-checked logical-consistency probe. On Qwen3-8B-Base, a task-specific self-judged DPO adaptation improves OntoNeg-WN by 5.27 percentage points but reduces ConceptShift by 37.50 points—despite humans reaching 82.7% and frontier LLMs 44–73% on OntoNeg—whereas mean change across five external accuracy suites is only -1.70 points. A development-tuned decoding rule partially mitigates ConceptShift regression to 12.93/14.60 points on the original/holdout splits while retaining most of the target gain. These results show that faithful post-training claims require joint reporting of gain, complementary abstraction regression, retention, drift, and equivalence—not proof that a single shared knowledge item degraded.

1 Introduction

Methods that adapt models without external or verifiable rewards—RLIF (Zhao et al., 2026), Self-Rewarding LM (Yuan et al., 2024), prompt-level self-refinement (Madaan et al., 2023)—are often

celebrated for single-task gains. Yet a method can raise accuracy on one knowledge-grounded probe while degrading complementary abstraction or logical consistency, drifting the language head, or exploiting a construction artifact. Suite-average retention can look nearly flat even when a local complementary axis collapses. This paper audits that failure mode with a conjunctive protocol and a structured-knowledge benchmark (Figure 1).

Running example. Consider the folk property “birds can fly.” *OntoNeg-WN* asks the model to prefer a same-category exception (penguin) over an unrelated counter-example (dog). *ConceptShift* asks it to recover the abstraction *bird* from a concrete pair (sparrow, penguin). *LogicRepair* is a synthetic SAT-checked probe that asks for minimal conflict repair in a Horn KB with the same surface motif. These are *structurally complementary* probes—not item-level paired triads over identical entities—so a large ConceptShift drop after OntoNeg-targeted training is evidence of cross-task complementary abstraction regression, not that one shared knowledge record locally degraded. We call target gain that hides such a complementary collapse a *sign-reversing local regression*.

Conjunctive protocol. The protocol reports five components (Table 27 in App. J): (i) *gain* on a designated target probe; (ii) *worst-task regression* on complementary probes; (iii) *retention* on external capability suites; (iv) *drift* of the predictive distribution (token-level KL); (v) *equivalence* (two-sided TOST) on complementary axes and *non-inferiority* on external retention.

SafeGain is the sole primary acceptance criterion:

target superiority plus complementary and retention non-inferiority. *LocalizedGain* (reported as AuditPass) is a secondary localization/equivalence diagnostic that additionally requires complementary-axis TOST under a pre-specified operational tolerance.

Findings. Comparing 8 methods on Qwen3-8B-Base ($n \geq 500$ /task; humans 82.7% / 85.3% / 76.4% and frontier LLMs 44–73% zero-shot on the same tasks):

1. A task-specific self-judged DPO adaptation of Self-Rewarding LM gains +5.27pp on OntoNeg-WN (Base 26.40%; chance 25%) but regresses −37.50pp on ConceptShift—a 42.77pp sign-reversal span—while mean external accuracy moves only −1.70pp. The pattern survives cluster-robust analysis, hard splits, ambiguity filtering, multi-gold scoring, and 8B cross-backbone replication. Development-tuned CSD partially mitigates ConceptShift regression to −12.93pp (holdout −14.60pp) while preserving most OntoNeg gain. Prompt-Dialectic instead loses −14.40pp via over-conservative refusal; Entropy-Min improves INA by +1.07pp but inflates OER by +9.93pp.
2. Conservative low-drift references (matched LoRA+KL and optional DION-Ref configurations) pass TOST at the ± 1 pp operational tolerance on every metric and calibrate the protocol when no target superiority is claimed.
3. Across 9 configuration-level run-terminal points, drift associates with worst-task regression ($\rho=0.72$). Certified-only OntoNeg retraining over 3 seeds (no kg_edge_absence rows) yields OntoNeg INA $44.27 \pm 0.76\%$ (+17.87pp vs. Base) with ConceptShift AbsAcc $16.10 \pm 0.17\%$ (−43.40pp; App. Table 26), so missing-edge labels are not required for the sign reversal.

Contributions. (1) DION-Bench, a 13,897-row stress test with attested-negation OntoNeg/ConceptShift evaluation items, a synthetic SAT-checked LogicRepair probe, and six validation checks (§3); (2) a conjunctive gain–regression–retention–drift–equivalence protocol with hierarchical uncertainty and multi-margin TOST (§4); (3) a task-specific self-judged DPO case study revealing a large sign-reversing complementary abstraction regression, with a decoding mitigation assessed on a post-tuning holdout (§5); (4) optional low-drift reference adapters

(LoRA+KL / DION-Ref) used only to instantiate and calibrate the protocol (§6).

2 Related Work

Internal feedback and self-rewarding post-training. Preference-based RLHF/DPO (Ouyang et al., 2022; Rafailov et al., 2023) and verifier-based RL (Shao et al., 2024; Guo et al., 2025) require external labels or oracles. Internal-feedback methods instead reuse the model as judge or confidence source: Self-Rewarding LM (Yuan et al., 2024), Intuitor (Zhao et al., 2026), process rewards from outcome labels (Yuan et al., 2025) or step-level verification (Lightman et al., 2023), and entropy-based intrinsic rewards. Zhang et al. (2025) document early gains followed by degradation under internal feedback; Trajectory-Mixed Supervision (Khan et al., 2026) links reward-free SFT to retention and KL drift. We do *not* propose another self-improving loop. We audit a task-specific self-judged DPO adaptation of the Self-Rewarding recipe on structured-knowledge probes and show that local complementary collapse can coexist with nearly flat suite-average retention.

Forgetting, alignment tax, and multi-capability evaluation. Alignment tax analyses report Pareto trade-offs between optimized objectives and pre-trained skills (Lin et al., 2024). Multifaceted forgetting suites after post-training likewise argue against single accuracy numbers (Thede et al., 2026). Relative to suite-average tax or capability averages, our protocol requires a *conjunctive* decision on a fixed target and complementary probes, with direction-aware non-inferiority and TOST equivalence, so a large local sign reversal cannot be cancelled by averaging.

Knowledge-grounded evaluation and faithfulness. Programmatic factuality and knowledge-grounded interactive evaluation construct or score claims against external knowledge (Speer et al., 2017; Miller, 1995; Han et al., 2024; Yu et al., 2024; Muhlgay et al., 2024). DION-Bench follows attested-evidence construction for OntoNeg/ConceptShift and couples complementary failure modes to a decision protocol rather than a single factuality score. LogicRepair is a synthetic SAT-checked logical-consistency probe (Ignatiev et al., 2018; Audemard et al., 2013)—not a third view of the same WordNet/ConceptNet items—so the suite measures cross-task structured-knowledge

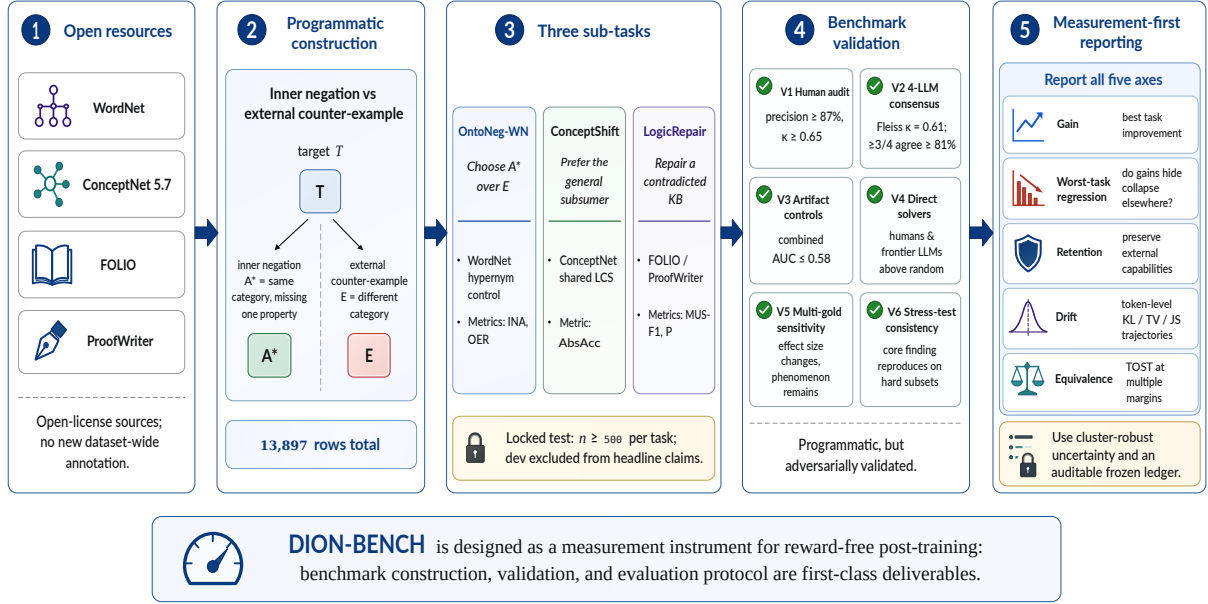


Figure 1: DION-Bench overview. **Finding:** single-probe gains can hide complementary abstraction regression. (1) WordNet 3.0 (Miller, 1995) and ConceptNet 5.7 (Speer et al., 2017) supply attested lexical evidence. (2) Programmatic construction separates inner negation A^* from external counter-example E . (3) Two lexical/ontological probes plus a synthetic logical-consistency probe. (4) Six validation checks (V1–V6), including human (82.7%) and frontier (44–73%) OntoNeg references. (5) Conjunctive reporting of gain, regression, retention, drift, and equivalence.

interference rather than paired-triad local knowledge rollback.

Conjunctive statistical reporting. Audit-Pass composes TOST equivalence (Schuirmann, 1987; Lakens, 2017), one-sided non-inferiority, intersection–union conjunction (Berger and Hsu, 1996), FDR control (Benjamini and Hochberg, 1995), and hierarchical bootstrapping into one reporting decision for reward-free post-training: which probes must improve, which must stay equivalent, and at which inferential level the decision is taken. Optional low-drift adapters (LoRA+KL; DION-Ref) serve only as calibrated references for that decision, not as the scientific center of the paper.

3 DION-Bench

DION-Bench comprises 13,897 programmatically constructed OntoNeg-WN, ConceptShift, and LogicRepair items; train/dev/test[conf] sizes appear with each sub-task below. *Construction sources* are Princeton WordNet 3.0 (Miller, 1995) (WordNet License) and the ConceptNet 5.7 dump of the open knowledge graph introduced by Speer et al. (2017) (paper describes ConceptNet 5.5; we use the publicly released 5.7 dump, CC BY-SA 4.0; attribution ships with the artifact). FOLIO and ProofWriter (Han et al., 2024; Tafjord et al., 2021)

are used only as *surface-style inspirations* for LogicRepair wording; no instance is taken from either corpus. Nodes use sense-preserving IDs (WordNet synset IDs; ConceptNet URIs retaining POS/sense suffixes when present) in a typed MultiDiGraph; construction is fully automatic with per-item provenance, and stratified human validation judgments (V1; §3.2) are released with the artifact package (App. I).

OntoNeg-WN and ConceptShift are lexical/ontological probes built from attested WordNet/ConceptNet evidence; LogicRepair is a separately synthesized Horn-chain logical-consistency probe with SAT certificates. The suite therefore tests complementary structured-knowledge capabilities—separating a same-category exception from an unrelated counter-example, recovering an abstraction, and identifying/repairing a synthetic logical conflict—rather than three views of the same item-level knowledge record.

Attested negation, not missing edges. Absence of an edge in a knowledge graph is not evidence that a property fails, so *no evaluation item* is built that way. Each evaluation item pairs two quotable pieces of lexical evidence under symmetric rules: an *affirmation* that some in-category member has the attribute, and a *denial* that A^* has it. Denials come from a WordNet gloss that lexically negates

the attribute (*tailless*, *without wings*) or from an explicit ConceptNet *Notr* assertion; affirmations come from a gloss asserting the same attribute outside any negating context, or from an attested attribute edge. Both quotations are released with the item. Reported scores use only these certified evaluation splits. The released OntoNeg-WN *training* pool additionally contains open-world heuristic rows tagged *kg_edge_absence* (1313/1597). Main-ledger checkpoints in Table 2 use that mixed pool for cross-method comparability; a certified-only retrain (284 rows; App. H) still exhibits OntoNeg gain with ConceptShift collapse, indicating the sign reversal is not an artifact of missing-edge supervision.

3.1 Sub-Tasks

OntoNeg-WN.

Train/dev/test/conf sizes are 1597/100/500/200. For a shared category S whose attribute (r, o) is affirmed by an in-category target T , the builder selects A^* within the bounded IsA subtree of S for which (r, o) is attested to fail, plus two in-category distractors that lack a matching denial in the released evidence (absence of denial is *not* treated as affirmative proof that the attribute holds; human auditors may mark such items invalid). A parent-disjoint E completes four candidates. Prompts name r and o in natural form (*does not have a tail*). *Metrics*: INA (accuracy of selecting A^*) and OER (rate of selecting E).

ConceptShift.

Train/dev/test/conf sizes are 4000/500/1000/500. Contrast pairs are taken only from attested WordNet antonym or ConceptNet Antonym/DistinctFrom edges, or from genuine co-hyponymy under a shared hypernym; relations are never reassigned to balance their frequencies. Both endpoints are resolved to WordNet senses, and S is a certified IsA-only least common subsumer whose ancestor paths $T \rightarrow S$ and $A \rightarrow S$ are at most four steps and are released with the item. Soft golds in *S_alternatives* are restricted to synset-verified synonyms and dominating parents, filtered for definitional fragments and offensive strings. Splits are concept-family-disjoint by S . *Metric*: AbsAcc—the single greedy generation is scored as a hit if it string-matches any accepted gold abstraction (S plus *S_alternatives*); it is not a model top- k list.

LogicRepair.

Train/dev/test sizes are 4000/500/1000. Each knowledge base is a *synthetic sentence-unit propositional* Horn chain—not derived from OntoNeg/ConceptShift entity IDs—with an injected contradiction a^* that negates one target sentence; the model must produce a minimal unsatisfiable subset (MUS) and a repaired KB $\mathcal{K}'$. Surfaces are styled after FOLIO and ProofWriter only. *Metrics*: The locked complementary decision axes use MUS precision (P) and MUS-F1 (macro-averaged independently against the released MUS gold; F1 is not recomputed from displayed aggregate P/R). MUS recall and $\mathcal{K}'$ exact-match are auxiliary repair diagnostics in App. Table 12 and are *not* part of SafeGain. All 1,000 test items carry SAT certificates for that gold (UNSAT + per-item MUS minimality; Appendix C). A CPU audit found no second deletion-minimal UNSAT subset on the test split under this encoding (0/1000); scoring remains relative to the released gold and is *not* a uniqueness theorem for all logically valid minimal repairs. Train and test are signature-disjoint.

Evaluation splits are concept-family-disjoint from training and from each other, with $n=500$ (OntoNeg-WN, capped by the supply of certified evidence) and $n=1,000$ (ConceptShift, LogicRepair). Each method is evaluated once on the test split; development splits are excluded from primary results. A separate holdout split (200 OntoNeg-WN and 500 ConceptShift rows, built after every decoding parameter was fixed) is reserved for §5. OntoNeg-WN has a four-option random-output reference (25%); ConceptShift and LogicRepair are open-generation tasks with no well-defined uniform chance level, so they use task-appropriate direct-solver and human references instead. Random-Negation is a training control, not a random-output baseline.

3.2 Benchmark Validation

Six independent validations establish benchmark quality. (V1) Stratified human audit of 1,120 rows (Table 1: per-task precision/ambiguity/ κ on that sheet alone; overall precision $\geq 87.0\%$). Separately, a 1,000-item ConceptShift re-audit yields majority accuracy 84.2%, $\kappa=0.72$, ambiguity 10.8% (App. E). Annotators judge natural-language plausibility from the released evidence quotations, not the graph paths, and may mark an item invalid. (V2) 4-LLM consensus over 6,000 judg-

ments, mean Fleiss $\kappa=0.61$, $\geq 3/4$ agreement $\geq 81\%$. (V3) Multivariate artifact controls: single-feature AUCs ≤ 0.55 , combined ≤ 0.58 , hard splits regress toward chance (App. D). (V4) Direct solvers: humans ($n=150/\text{task}$) reach 82.7% / 85.3% / 76.4%; frontier LLMs 44–73% zero-shot (Table 24). (V5) The Self-Rewarding regression holds from single-gold (−20.8pp) to ancestor-permissive scoring (−36.4pp; App. G). (V6) Findings reproduce on hard splits, specificity-controlled pairs, and the post-tuning holdout within $\pm 20\%$ effect sizes.

Sub-task	Prec.↑ human	Amb.↓ human	$\kappa_{\text{Coh}}\uparrow$ 3-rater	$\kappa_{\text{F1}}\uparrow$ 4-LLM
OntoNeg-WN ($n=500$)	87.0%	9.6%	0.66	0.64
ConceptShift ($n=400$)	87.0%	11.5%	0.66	0.59
LogicRepair ($n=220$)	90.5%	9.5%	0.71	0.61
<i>Artifact-control AUC (§3.2 V3)</i>				
length	freq.	graph-deg.	position	overlap
0.52	0.54	0.51	0.50	0.55

Table 1: DION-Bench validation: human audit (V1, 1,120 rows of the released stratified audit; mean pairwise Cohen κ over three raters), LLM consensus (V2, 6,000 judgments; Fleiss κ in the last column), and artifact controls (V3, every single-feature AUC ≤ 0.55). Quality bars are met on all three sub-tasks, and (O1) reproduces on the DION-Bench-Core split ($n=2,220$: certified OntoNeg-WN and LogicRepair items plus the high-confidence rows of the ConceptShift re-audit).

4 Experiments

4.1 Setup and Evaluation Protocol

We use Qwen3-8B-Base (Yang et al., 2025) as the primary backbone with 4-bit QLoRA (Hu et al., 2022; Detrmers et al., 2023) on 2×RTX 4090D, and validate cross-backbone transfer on Llama-3.1-8B (Grattafiori et al., 2024) and Qwen3-8B-Instruct (same 8B scale). The comparison covers 8 methods—Random-Negation, Prompt-Dialectic (Madaan et al., 2023), Entropy-Min, Self-Rewarding LM (Yuan et al., 2024), Self-Critique, SFT-LoRA, LoRA+KL, and DION-Ref—plus Base; DION-Ref contributes multiple hyperparameter rows in Table 2 but counts as one method for the 8-method ranking. Additional mechanism ablations appear in §4.6. Evaluation uses greedy decoding, $n\geq 500$, 3–7 seeds, and validated per-item predictions (App. C).

4.2 Statistical Protocol

Primary decisions use a *two-way hierarchical bootstrap* (Efron, 1979): training runs are resampled

with replacement, then concept-family clusters within each drawn run, so an interval reflects run-to-run and item-to-item variation jointly; item- and cluster-level bootstraps are narrower supporting views (§4.4), never the decision basis. On top we apply TOST at four margins (± 0.5 , ± 1 , ± 1.5 , $\pm 2\text{pp}$; App. B), with clusters drawn once per replicate and shared by both arms, and Benjamini–Hochberg correction over discovery tests.

We fix $\pm 1\text{pp}$ as a *pre-specified operational tolerance* for TOST (protocol card; not a domain indifference region or public preregistration). MDE at 80% power spans 0.30–0.98pp (widest two-way half-width 0.65pp on OER), so axes resolve below $\pm 1\text{pp}$ here, but resolvability does not justify the scientific size of δ . Lower-is-better probes use directed NI (OER +1pp; WikiText-PPL +5% relative).

Reporting decisions. *SafeGain* is the sole primary acceptance rule (target superiority plus complementary and retention NI). *LocalizedGain/AuditPass* is a secondary TOST localization diagnostic ($|\Delta| < \delta$; collateral gains also fail). BH applies to discovery tests only. The constructive positive control *gain_matched_safe* shows the rules are reachable (not an independent SOTA method). RetentionNI without per-seed SDs uses modeled uncertainty and is weaker than empirical TOST. LogicRepair enters via MUS-P/MUS-F1; MUS-R and $\mathcal{K}'$ exact are auxiliary (App. Table 12). Training prompts never include gold answers.

4.3 Primary Results

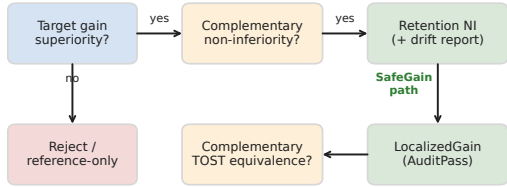
Finding 1: sign-reversing local regression. Self-Rewarding gains +5.27pp INA on OntoNeg-WN but regresses −37.50pp on ConceptShift (Figure 3). Humans reach 82.7% and frontier LLMs 44–73% on OntoNeg (Table 24; Figure 6), so the +5.27pp lift is not merely noise around an unsolvable task; yet mean external retention moves only −1.70pp, so averaging hides the local failure.

Finding 2: partial mitigation without erasing the audit need. Development-tuned CSD partially mitigates ConceptShift regression to −12.93pp while preserving OntoNeg gain (+4.13pp vs. +5.27pp); the post-tuning holdout yields −14.60pp (§5). Residual collapse remains large, so CSD does not “recover” the complementary axis. Prompt-Dialectic loses −14.40pp INA via over-conservative refusal; Self-Critique shows a smaller same-direction pattern (+2.87/−14.17pp).

Method	OntoNeg-WN		ConceptShift AbsAcc↑ <i>n</i> =1,000	LogicRepair	
	INA↑ <i>n</i> =500	OER↓		P↑ <i>n</i> =1,000	F1↑ <i>n</i> =1,000
Base (Qwen3-8B-Base)	26.40	20.20	59.50	62.85	44.98
Random-Negation	25.67±0.83	28.47±0.90	24.80±1.11	29.26±0.88	30.35±0.78
△ vs Base	−0.73	+8.27	−34.70	−33.59	−14.62
Prompt-Dialectic	12.00	5.20	34.30	48.30	36.17
△ vs Base	−14.40	−15.00	−25.20	−14.55	−8.81
Entropy-Min	27.47±0.83	30.13±1.01	52.40±1.01	58.33±0.75	42.13±0.66
△ vs Base	+1.07	+9.93	−7.10	−4.52	−2.85
Self-Critique	29.27±0.90	18.80±0.72	45.33±0.97	61.08±0.67	43.93±0.59
△ vs Base	+2.87	−1.40	−14.17	−1.77	−1.04
Self-Rewarding	31.67±0.90	11.67±0.70	22.00±1.01	50.88±0.81	43.62±0.66
△ vs Base	+5.27	−8.53	−37.50	−11.97	−1.36
SFT-LoRA (control)	25.93±0.81	21.47±0.83	61.30±0.92	63.16±0.55	45.45±0.49
LoRA+KL (matched control)	26.40±0.55	20.36±0.62	59.62±0.75	62.86±0.46	44.83±0.41
<i>DION-Ref configuration sweep, mean±s.d. over training runs</i>					
SuperCon ($\lambda_P=10$, $r=2$, 250 step, 0.35M, 7 runs)	26.46±0.36	20.34±0.38	59.69±0.43	62.85±0.28	44.76±0.26
△ vs Base	+0.06	+0.14	+0.19	+0.00	−0.22
TRA-only ($\lambda_P=5$, $r=4$, 500 step, 1.9M, 5 runs)	26.28±0.48	20.36±0.57	59.70±0.62	62.95±0.42	44.89±0.39
△ vs Base	−0.12	+0.16	+0.20	+0.10	−0.08
Mid ($\lambda_P=3$, $r=8$, 800 step, 7.7M, 7 runs)	26.37±0.53	20.43±0.56	59.19±0.67	62.97±0.42	44.89±0.39
△ vs Base	−0.03	+0.23	−0.31	+0.12	−0.09
Full-LoRA ($\lambda_P=2$, $r=16$, 1500 step, 43.6M, 4 runs)	18.90±2.94	24.85±3.35	52.23±3.88	56.23±2.38	42.04±2.10
△ vs Base	−7.50	+4.65	−7.27	−6.62	−2.93

Table 2: Primary results on DION-Bench (greedy decoding, BH-corrected p ; every cell recomputed from per-item predictions). LogicRepair P/F1 are macro-averaged independently (F1 is not recomputed from displayed aggregate P/R); MUS-R and K' exact-match are in App. Table 12. Deltas use unrounded values from the released result ledger and may differ by 0.01pp from subtraction of displayed cells. The table lists Base plus method/config rows; the 8-method ranking treats DION-Ref as one method. “Self-Rewarding” is the task-specific self-judged DPO adaptation (+5.27/−37.50pp).

Conjunctive protocol decision flow



Report all five axes even when the conjunctive decision rejects.

Figure 2: Conjunctive decision flow. SafeGain is the primary acceptance path; LocalizedGain/AuditPass is a secondary TOST localization diagnostic. Drift is always reported.

Finding 3: conjunctive decisions reject the gain. The self-judged DPO adaptation clears target superiority but fails SafeGain and the LocalizedGain/AuditPass diagnostic (Table 3; full matrix in App. Table 6). GainMatched-Safe is a constructive positive control under modeled external-retention uncertainty; LoRA+KL and conservative DION-Ref are low-drift references without target superiority. Three of 8 methods meet target superiority alone but not the AuditPass diagnostic—Self-Rewarding, Entropy-Min, Self-Critique—and OntoNeg-gain rank versus worst-task rank has

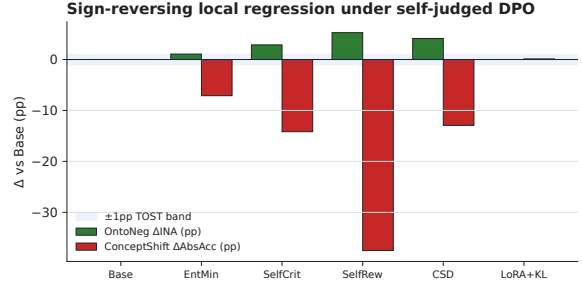


Figure 3: Signed deltas vs. Base (pp). Self-judged DPO yields OntoNeg gain with ConceptShift collapse; CSD partially mitigates the regression; LoRA+KL stays near the ± 1 pp operational tolerance. Extended drift diagnostics: Figure 8.

Spearman $\rho = -0.14$.

Finding 4: certified-only multi-seed sensitivity. Retraining on the 284-row certified-only OntoNeg pool (no `kg_edge_absence`) over 3 seeds yields INA $44.27 \pm 0.76\%$ and ConceptShift AbsAcc $16.10 \pm 0.17\%$ (+17.87/−43.40pp vs. Base; App. Table 26, Figure 4). The same 3-seed protocol for Self-Critique, Entropy-Min, SFT-LoRA, Shuffled-DPO, and LoRA+KL reproduces the sign reversal (INA 44.27–52.33%; AbsAcc 14.03–15.73%). The mixed-pool ledger in Table 2 remains

Method	TargetSup	CompTOST	RetNI	AuditPass
Self-Rewarding	✓	✗	✗	✗
Self-Critique	✓	✗	✗	✗
Entropy-Min	✓	✗	✗	✗
LoRA+KL	✗	✓	✓	✗
GainMatched-Safe	✓	✓	✓	✓

Table 3: Compact conjunctive decisions (full matrix: App. Table 6). Self-judged gains fail AuditPass; GainMatched-Safe is a modeled control.

the multi-method comparison; certified-only isolates the training-label confounder (training-mix schematic: Figure 7).

All three conservative DION-Ref configurations and matched LoRA+KL pass TOST at ± 1 pp on all five metrics under the hierarchical, cluster and item bootstraps alike (Table 7 in App. B). At ± 0.5 pp only LogicRepair F1 survives for every reference (SuperCon also on LogicRepair P; LoRA+KL also on INA, LogicRepair P); the other axes sit below the noise floor there (Table 5 in App. B.2). The full AuditPass decision, including directed external retention NI and relative PPL NI (upper-bound for lower-is-better), appears in App. Table 6.

Full-LoRA drifts to $\bar{D}_{KL}=0.18$ nats and regresses -7.27 pp on ConceptShift (Figure 8, right). Across 9 configuration-level run-terminal points, drift is associated with worst-task regression at Spearman $\rho=0.72$ (Table 17); nested checkpoint trajectories are not independent samples. Drift tracks risk scale; mechanism is discussed in §5.

4.4 Cluster-Robust Uncertainty

Programmatic items share target concepts, hypernyms, and source relations, so an item-level bootstrap can understate uncertainty. Recomputing the ConceptShift effect under 4 resampling units (Table 13 in App. D) moves interval widths by at most 9%—these families do not group enough items for cluster resampling to matter, which is why the primary level resamples runs too. SuperCon stays inside ± 1.5 pp under every unit; the regression stays below -33.9 pp at its most favourable bound and reads -38.2 pp when families are weighted equally.

4.5 Cross-Backbone Diagnostic Validity

A two-tier study at the same 8B scale (Table 19 in App. F) covers inference-only evaluation on Llama-3.1-8B and Qwen3-8B-Instruct ($n=500$ per task on DION-Bench-Core) and a small-budget post-training replication (≤ 500 steps, 3 runs). Absolute scores vary by family and alignment, but the gain/regression pattern persists on all three backbones, retaining 61% (Llama) / 52% (Instruct) of

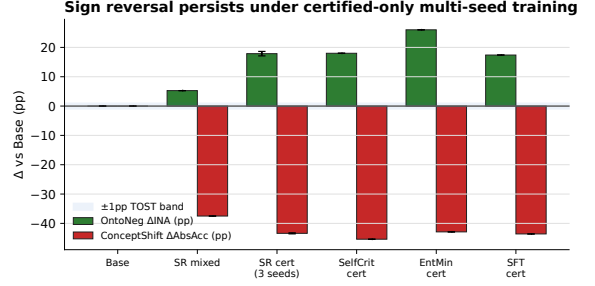


Figure 4: Signed deltas vs. Base under certified-only Self-Rewarding (3 seeds; error bars = s.d.): both OntoNeg gain and ConceptShift collapse enlarge relative to the mixed-pool ledger; multi-seed controls share the direction (App. Table 26).

the Qwen3-8B-Base effect size with cluster CIs excluding zero.

4.6 Baseline Fidelity, Retention, and Reference Quality

Self-Rewarding fidelity is stable across 18 training-time variants ($+3.8$ – $+6.9$ pp ON; -31 – -42 pp CS) plus $+8.7$ pp on a held-out instruction-following task; Self-Critique shows the same direction ($+2.87$ – -14.17 pp; App. C). On external suites (GSM8K (Cobbe et al., 2021), MMLU-Pro (Wang et al., 2024), HellaSwag (Zellers et al., 2019), ARC-C (Clark et al., 2018), TruthfulQA (Lin et al., 2022), WikiText-2 PPL (Merity et al., 2017)), DION-Ref stays within $|\bar{\Delta}| \leq 0.17$ pp while Self-Rewarding’s mean accuracy Δ is only -1.70 pp; RetentionNI without per-seed seed_sd uses modeled SDs (Table 21; App. Table 6). Ablations under TRA-only show projector removal raises dispersion/drift, while boundary/single-branch changes stay near noise; LoRA+KL matches ± 1 pp TOST but has 71% higher 95th-percentile KL (Tables 22, 23; Fig. 8).

5 Analysis

5.1 Over-Specific Answers Dominate ConceptShift Errors

ConceptShift asks for the least common subsumer, yet Self-Rewarding errors concentrate on over-specific candidates (52% vs. 5% for Base; Figure 5); on unanimously correct items the regression remains -36.6 pp ($n=724$). A specificity-score model separates preferred from rejected answers at AUC= 0.72 (OR= 1.59), and specificity-swapped pairs reverse the self-judge but not Base (App. G, E). Because OntoNeg training mixes certified denials with kg_edge_absence heuristics, we treat specificity bias as an error-mode-consistent account

rather than an exclusive cause.

5.2 One Frozen Decoding Rule Partially Mitigates Both Selection Tasks

Context-anchored specificity-penalized decoding (CSD) re-ranks prompt-visible beam candidates without gold, S , or paths. When parent records carry candidate_logprobs, those scores are used; otherwise the released intervention uses a documented parent-logprob/rank/length proxy (reproducible but heuristic). Dev-frozen coefficients transfer to test and the post-tuning hold-out: ConceptShift regression falls from -37.50pp to -12.93pp (OntoNeg $+4.13\text{pp}$ vs. $+5.27\text{pp}$; App. Table 9) with large residual failure (App. E.3).

5.3 Drift Scales Risk; Objectives Set Direction

Drift associates with regression *magnitude* while the training objective tracks *direction* (Table 17); label breadth changes magnitude (-20.8 – -36.4pp) but not existence. The protocol therefore reports target superiority, complementary equivalence, and retention NI jointly rather than averaging.

6 Low-Drift Reference Adapters

When a method claims no target superiority, we calibrate with matched LoRA+KL and optional DION-Ref anchors (App. A). Conservative points (SuperCon, TRA-only, Mid) and LoRA+KL clear $\pm 1\text{pp}$ TOST on all five axes and separate on tails (App. Tables 23, 4); Full-LoRA is a high-capacity outlier. Ablations that strip TRA/boundary terms while keeping KL/projection often stay near Base, so we treat these adapters as protocol instruments, not primary contributions (§4.6).

7 Discussion and Conclusion

A single grounded gain is not faithful improvement: $+5.27\text{pp}$ on OntoNeg can coexist with -37.50pp on ConceptShift while mean external accuracy moves only -1.70pp ; certified-only multi-seed re-training enlarges the pattern ($+17.87$ – -43.40pp). Because the probes are not paired triads, we read complementary abstraction regression rather than proven rollback of one shared knowledge record. Task-specific self-judged DPO is high-risk for that failure; CSD and low-drift references partially mitigate but do not replace conjunctive rejection of sign-reversing complementary regressions.

Limitations

Scale and construct scope. We study an English 8B greedy case (Qwen3-8B-Base; same-scale Llama/Instruct checks). Larger models, multilingual settings, and interactive multi-turn grounding are out of scope. OntoNeg and ConceptShift are not item-level paired triads; LogicRepair is synthetic Horn-chain logic rather than the same WordNet/ConceptNet facts, so we claim complementary abstraction interference—not local rollback of one shared knowledge record. Paired-triad construction that would support a rollback reading remains future work. Human audits cover stratified samples rather than every released item; residual labeling disagreement is reported in App. I.

Labels, seeds, and retention. Mixed-pool OntoNeg training includes kg_edge_absence (1313/1597); evaluation is certified-only. Certified-only Self-Rewarding (284 rows; 3 seeds) still yields INA $44.27 \pm 0.76\%$ and ConceptShift AbsAcc $16.10 \pm 0.17\%$; Self-Critique, Entropy-Min, SFT-LoRA, Shuffled-DPO, and LoRA+KL match the direction under the same 3-seed protocol. RetentionNI without empirical per-seed SDs uses modeled uncertainty and is weaker than run-aware TOST; full external-suite per-seed harnesses remain incomplete, so we do not treat modeled RetentionNI as a substitute for measured seed dispersion. Where seed-level external predictions are missing, we report the modeled interval explicitly rather than implying empirical dispersion.

Protocol, metrics, and CSD. The $\pm 1\text{pp}$ TOST gate is a pre-specified operational tolerance on the protocol card, not a domain indifference region or public preregistration. SafeGain is primary; LocalizedGain/AuditPass is a localization diagnostic. LogicRepair enters via MUS-P/F1; MUS-R and $\mathcal{K}'$ exact are auxiliary. Released mixed-pool CSD numbers use a documented parent-logprob/rank/length proxy when per-candidate likelihoods are absent; certified- and mixed-parent candidate-logprob regenerations are released for audit, but order-shuffle and coverage checks remain open. DION-Ref and LoRA+KL are protocol instruments rather than standalone SOTA methods. Cross-backbone checks use a small post-training budget and should not be read as exhaustive transfer. We do not claim production readiness or that conjunctive rejection prevents all post-training regression outside this English 8B greedy case. Ethics: App. I.

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References

- Gilles Audemard, Jean-Marie Lagniez, and Laurent Simon. 2013. Improving Glucose for incremental SAT solving with assumptions: Application to MUS extraction. In *Theory and Applications of Satisfiability Testing – SAT 2013*, pages 309–317. Springer.
- Yoav Benjamini and Yosef Hochberg. 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B (Methodological)*, 57(1):289–300.
- Roger L. Berger and Jason C. Hsu. 1996. Bioequivalence trials, intersection-union tests and equivalence confidence sets. *Statistical Science*, 11(4):283–319.
- Andrei Z. Broder. 1997. On the resemblance and containment of documents. In *Proceedings of the Compression and Complexity of Sequences*, pages 21–29. IEEE.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *Preprint*, arXiv:1803.05457.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *Preprint*, arXiv:2110.14168.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient fine-tuning of quantized LLMs. In *Advances in Neural Information Processing Systems*, volume 36, pages 10088–10115.
- Bradley Efron. 1979. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. 2024. The Llama 3 herd of models. *Preprint*, arXiv:2407.21783.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. 2025. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645(8081):633–638.
- Simeng Han, Hailey Schoelkopf, Yilun Zhao, Zhen-qing Qi, Martin Riddell, Wenfei Zhou, James Coady, David Peng, Yujie Qiao, Luke Benson, Lucy Sun, Alexander Wardle-Solano, Hannah Szabó, Ekaterina Zubova, Matthew Burtell, Jonathan Fan, Yixin Liu, Brian Wong, Malcolm Sailor, et al. 2024. FOLIO: Natural language reasoning with first-order logic. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 22017–22031, Miami, Florida, USA. Association for Computational Linguistics.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.
- Alexey Ignatiev, Antonio Morgado, and Joao Marques-Silva. 2018. PySAT: A Python toolkit for prototyping with SAT oracles. In *Theory and Applications of Satisfiability Testing – SAT 2018*, pages 428–437. Springer.
- Rana Muhammad Shahroz Khan, Zijie Liu, Zhen Tan, Charles Fleming, and Tianlong Chen. 2026. TMS: Trajectory-mixed supervision for reward-free, on-policy SFT. *Preprint*, arXiv:2602.03073.
- Daniël Lakens. 2017. Equivalence tests: A practical primer for *t* tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2023. Let’s Verify Step by Step. *Preprint*, arXiv:2305.20050.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252, Dublin, Ireland. Association for Computational Linguistics.

- Yong Lin, Hangyu Lin, Wei Xiong, Shizhe Diao, Jianmeng Liu, Jipeng Zhang, Rui Pan, Haoxiang Wang, Wenbin Hu, Hanning Zhang, Hanze Dong, Renjie Pi, Han Zhao, Nan Jiang, Heng Ji, Yuan Yao, and Tong Zhang. 2024. Mitigating the alignment tax of RLHF. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 580–606, Miami, Florida, USA. Association for Computational Linguistics.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-Refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, volume 36, pages 46534–46594.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2017. Pointer sentinel mixture models. In *International Conference on Learning Representations*.
- George A. Miller. 1995. WordNet: A lexical database for English. *Communications of the ACM*, 38(11):39–41.
- Dor Muhlgay, Ori Ram, Inbal Magar, Yoav Levine, Nir Ratner, Yonatan Belinkov, Omri Abend, Kevin Leyton-Brown, Amnon Shashua, and Yoav Shoham. 2024. Generating benchmarks for factuality evaluation of language models. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 49–66, St. Julian’s, Malta. Association for Computational Linguistics.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F. Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, pages 53728–53741.
- Donald J. Schuirmann. 1987. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6):657–680.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *Preprint*, arXiv:2402.03300.
- Robyn Speer, Joshua Chin, and Catherine Havasi. 2017. ConceptNet 5.5: An open multilingual graph of general knowledge. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 31.
- Oyvind Tafjord, Bhavana Dalvi, and Peter Clark. 2021. ProofWriter: Generating implications, proofs, and abductive statements over natural language. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 3621–3634, Online. Association for Computational Linguistics.
- Lukas Thede, Stefan Winzeck, Zeynep Akata, and Jonathan Richard Schwarz. 2026. CapTrack: Multi-faceted evaluation of forgetting in LLM post-training. *Preprint*, arXiv:2603.06610.
- Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhramil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. 2024. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Advances in Neural Information Processing Systems*, volume 37, pages 95266–95290.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chuji Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, et al. 2025. Qwen3 technical report. *Preprint*, arXiv:2505.09388.
- Zhuohao Yu, Chang Gao, Wenjin Yao, Yidong Wang, Wei Ye, Jindong Wang, Xing Xie, Yue Zhang, and Shikun Zhang. 2024. KIEval: A knowledge-grounded interactive evaluation framework for large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5967–5985, Bangkok, Thailand. Association for Computational Linguistics.
- Lifan Yuan, Wendi Li, Huayu Chen, Ganqu Cui, Ning Ding, Kaiyan Zhang, Bowen Zhou, Zhiyuan Liu, and Hao Peng. 2025. Free process rewards without process labels. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 73511–73525. PMLR.
- Weizhe Yuan, Richard Yuanzhe Pang, Kyunghyun Cho, Xian Li, Sainbayar Sukhbaatar, Jing Xu, and Jason E. Weston. 2024. Self-Rewarding language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 57905–57923. PMLR.
- Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. HellaSwag: Can a machine really finish your sentence? In *Proceedings of*

the 57th Annual Meeting of the Association for Computational Linguistics, pages 4791–4800, Florence, Italy. Association for Computational Linguistics.

Yanzhi Zhang, Zhaoxi Zhang, Haoxiang Guan, Yilin Cheng, Yitong Duan, Chen Wang, Yue Wang, Shuxin Zheng, and Jiyan He. 2025. No Free Lunch: Rethinking internal feedback for LLM reasoning. *Preprint*, arXiv:2506.17219.

Xuandong Zhao, Zhewei Kang, Aosong Feng, Sergey Levine, and Dawn Song. 2026. Learning to reason without external rewards. In *International Conference on Learning Representations*, pages 2548–2581.

A Proofs and Method Details

A.1 Proofs

Proposition 1 (Initialization Identity). *Let $\mathcal{F}_0$ and $\mathcal{F}_{\text{DION-Ref}}$ be the function classes of the base and DION-Ref-augmented models. Then $\mathcal{F}_0 \subseteq \mathcal{F}_{\text{DION-Ref}}$, and zeroing the TRA residual path—setting the branch outputs $b_{1,\ell}=b_{2,\ell}=0$ and the third-branch maps $U_3=0$ (equivalently $s_\ell=0$)—is sufficient for $\pi_{\text{DION-Ref}}=\pi_{\theta_0}$ at initialization; other equivalent zeroings may exist, so this is not claimed as necessary. Empirically, with all residual outputs zeroed, $\max_x \|\pi_\theta(\cdot|x)-\pi_{\theta_0}(\cdot|x)\|_\infty=0$ to fp32 precision.*

With $b_{1,\ell}=b_{2,\ell}=0$ and $U_3=0$, the TRA branch outputs $s_\ell=0$ identically; thus $\Delta H_\ell=0$ and $\hat{H}_\ell=H_\ell^0$ exactly. The output distribution of the DION-Ref-augmented model coincides with the base.

Proof sketch of Proposition 2. For any x , Pinsker’s inequality gives $\text{TV}(\pi_\theta\|\pi_{\theta_0}) \leq \sqrt{D_{\text{KL}}/2}$. Taking expectation over $x \sim \mathcal{D}$ and applying Jensen: $\mathbb{E}_{\mathcal{D}} \text{TV} \leq \sqrt{\mathbb{E}_{\mathcal{D}} D_{\text{KL}}/2} \leq \sqrt{\epsilon/2}$.

Local Tension Descent (sanity property). By smoothness, $B(c+\Delta) \leq B(c) + \nabla B^\top \Delta + \frac{L}{2} \|\Delta\|^2$. Substituting $\Delta = -\alpha P \nabla B$ yields $B(c^+) \leq B(c) - \alpha(1 - \frac{L\alpha}{2}) \|P \nabla B\|^2$.

Unique minimizer of $\mathcal{L}_{\text{ABO}}$ on a fixed-feature simplex (sanity property). Holding the codebook features fixed and optimizing only the simplex mixture q , $D_{\text{KL}}(\cdot\|q_T)$ is strictly convex on the open simplex and $\lambda_C \|q - u\|^2$ adds strong convexity, so that restricted subproblem has a unique minimizer. This does not claim uniqueness of the full TRA/adaptor parameterization.

A.2 Architecture and Training Details

The Stage-1 Concept Predicate Head (CPH) is a linear codebook $q_\phi(z | c_\ell) = \text{softmax}(W_z c_\ell)$ over $K=512$ latent codes, trained self-supervisedly with

two stochastic views and a batch-entropy regulariser:

$$\mathcal{L}_{\text{CPH}} = D_{\text{JS}}(q_\phi^{(1)}\|q_\phi^{(2)}) - \lambda_{\text{div}} H(\bar{q}_\phi). \quad (1)$$

The boundary energy is defined as:

$$B_\ell(c_\ell) = \frac{H_\phi(c_\ell)}{\log K} + \beta \text{Var}_{z \sim q_\phi}[s_z(c_\ell)]. \quad (2)$$

Let $\pi_{\theta_0}(y | x)$ be a frozen base LM with hidden states $H_\ell(x) \in \mathbb{R}^{n \times d}$ at layer ℓ . We pool H_ℓ via attention pooling $c_\ell = \sum_t a_{\ell,t} h_{\ell,t}$. For Qwen3-8B-Base ($d=4096$, 36 layers) we inject DION-Ref at $\mathcal{L}=\{12, 24, 35\}$ with mixing weights $\omega=(0.25, 0.35, 0.40)$.

The direction operator estimates a *tractable surrogate* of predictive sensitivity of $\pi_\theta(\cdot | x)$ w.r.t. the pooled concept state c_ℓ : $m=4$ Monte-Carlo tokens are drawn from cached next-token log-probabilities (detached sampling distribution), mapped through a fixed Gaussian concept-to-vocabulary projection $R_\ell \in \mathbb{R}^{d \times r}$ ($r=64$) with an LM-head-aligned fallback when caches are unavailable. This is *not* the full downstream Transformer Jacobian $\nabla_c \log \pi_\theta(y | x)$; we treat it as an LM-head-aligned sensitivity proxy and form the low-curvature projector $P_{\ell,\epsilon} = U_{\ell,\epsilon} U_{\ell,\epsilon}^\top$ over eigendirections with $\lambda_i \leq \epsilon \lambda_r$, $\epsilon=0.03$:

$$\delta_\ell(c) = \frac{R_\ell P_{\ell,\epsilon} R_\ell^\top \nabla_c B_\ell(c)}{\|R_\ell P_{\ell,\epsilon} R_\ell^\top \nabla_c B_\ell(c)\|_2 + \delta}, \quad (3)$$

i.e. a low-surrogate-curvature, high-boundary-disagreement direction. At each $\ell \in \mathcal{L}$, the triadic residual adaptor (TRA) adds:

$$s_\ell = g_1 \odot b_{1,\ell} + g_2 \odot b_{2,\ell} + g_3 \odot U_3 \sigma(V_3 [c; \delta; c \odot \delta]), \quad (4)$$

broadcast to tokens via $\Delta h_{\ell,t} = a_{\ell,t} s_\ell$ with weight α_{TRA} .

Proposition 2 (Pinsker Drift Bound on the Evaluation Distribution). *For evaluation distribution $\mathcal{D}$, if $\mathbb{E}_{x \sim \mathcal{D}} D_{\text{KL}}(\pi_\theta(\cdot|x)\|\pi_{\theta_0}(\cdot|x)) \leq \epsilon$, then $\mathbb{E}_{x \sim \mathcal{D}} \text{TV}(\pi_\theta, \pi_{\theta_0}) \leq \sqrt{\epsilon/2}$.*

With perturbed $c_A = c + \rho \delta$, sublated $c_S = c + s_\ell$, and $q_{T,A,S} = q_\phi(\cdot | c_{T,A,S})$, the full ABO objective is:

$$\begin{aligned} \mathcal{L}_{\text{ABO}} = & \lambda_P D_{\text{KL}}(\pi_\theta\|\pi_{\theta_0}) + \lambda_A D_{\text{KL}}(q_A\|q_S) \\ & + \lambda_R [\mathcal{T}_\ell(c_S) - \mathcal{T}_\ell(c) + m_T]_+ \\ & + \lambda_C (H(q_S) - \kappa \log K)^2. \end{aligned} \quad (5)$$

The four terms enforce: preserving language ability via KL, absorbing the boundary-tension direction, reducing tension, and avoiding vacuous uniformity.

B Hyperparameters and Statistical Details

B.1 Compute Budget

Total compute ≈ 48 GPU-hours on $2 \times \text{RTX } 4090\text{D}$ (job breakdown in the released compute ledger) plus a 5,000-call API budget for audits. V2 LLM consensus covers 500 items per task across three tasks with four judges ($500 \times 3 \times 4 = 6,000$ item-judge judgments; product-family APIs). Separately, the released API-call ledger records billed API calls: 2,400 frontier direct-solve calls (600 items $\times$ 4 models), 300 batched judge calls (600 items, batch size 8), and 800 intervention reruns, totaling 3,500 against a 5,000 cap. Judgment counts and call counts are not interchangeable. Cross-backbone Tier-A inference: <1 GPU-hour; Tier-B post-training: <12 GPU-hours.

Config	rank	Targets	λ_P	Steps	Params	Runs
SuperCon	2	Q,V	10	250	0.35M	7
TRA-only	4	Q,V	5	500	1.9M	5
Mid	8	Q,K,V,O	3	800	7.7M	7
Full-LoRA	16	all 7	2	1,500	43.6M	4
LoRA+KL	4	Q,V	5	500	1.9M	5
SFT-LoRA	4	Q,V	—	500	1.9M	3

Table 4: Configuration-level hyperparameters, read from the released run configurations and prediction records (**Runs** counts distinct training seeds). All runs use $\alpha_{\text{TRA}}=0.05$, 4-bit QLoRA, batch 32, and lr 1×10^{-5} except Full-LoRA (3×10^{-5}).

B.2 Statistical Power and Equivalence Sensitivity

Run budgets are listed per configuration in Table 4: conservative DION-Ref configurations and the matched LoRA+KL control use 5–7 training runs, gain-seeking baselines 3–4. All primary equivalence decisions come from the two-way hierarchical bootstrap (runs, then concept-family clusters); item-level and cluster-level bootstraps are reported alongside as narrower views. Relative to the item-level interval (the released interval-width summary), the cluster-level interval is $0.95\text{--}1.16\times$ as wide and the two-way interval $1.13\text{--}2.63\times$: cluster resampling only widens ConceptShift, where concept families group many items (up to $1.16\times$), and leaves LogicRepair, OntoNeg-WN essentially unchanged ($\leq 1.05\times$), whereas resampling training runs widens every axis. That is the concrete reason primary decisions use the two-way level rather than a cluster-robust item bootstrap.

The released equivalence-margin summary records, per metric, the across-run SD, the observed

two-way half-width $((hi-lo)/2)$, the minimum effect resolvable at 80% power for two one-sided 5% tests, and the number of runs that halving the margin would need: MUS-F1 0.30pp (3); INA 0.76pp (17); MUS-P 0.84pp (20); AbsAcc 0.86pp (21); OER 0.98pp (28). The quantities use a released equivalence-margin procedure whose unit of inference is the training run, so items never multiply the effective sample size. Under the corrected half-width, all five axes (MUS-F1, INA, MUS-P, AbsAcc, OER) resolve an effect strictly below the $\pm 1\text{pp}$ band at the present budget; we treat $\pm 1\text{pp}$ as a fixed protocol tolerance rather than a domain-derived indifference region.

We report TOST at $\pm 0.5\text{pp}$ (stringent), $\pm 1\text{pp}$ (protocol margin), $\pm 1.5\text{pp}$, and $\pm 2\text{pp}$ (noise-aware). At $\pm 1\text{pp}$ every conservative reference (SuperCon, TRA-only, Mid, LoRA+KL) passes on all five metrics at the hierarchical, cluster, and item levels. At $\pm 0.5\text{pp}$ only LogicRepair F1 passes for all four (SuperCon also on LogicRepair P; LoRA+KL also on INA, LogicRepair P); every other axis lies below the noise floor at that margin. Table 8 repeats the decision on two audited subsets: removing every annotator-flagged item leaves the references equivalent at $\pm 1\text{pp}$, while the smaller Core subset raises the worst reference half-width enough to need $\pm 1.5\text{pp}$ —a resolution effect, since no reference point estimate moves materially. Gain-seeking methods fail at every margin on every subset, so the qualitative conclusion is margin-invariant. We further distinguish behavioral task equivalence (under noisy labels) from distributional low drift (KL/TV/JS): the latter is independently established (Table 16).

Config	$\pm 1\text{pp}$, all five axes			$\pm 0.5\text{pp}$ axes passing
	hier.	clust.	item	
SuperCon	✓	✓	✓	P, F1
TRA-only	✓	✓	✓	F1
Mid	✓	✓	✓	F1
LoRA+KL	✓	✓	✓	INA, P, F1

Table 5: Equivalence against Base at the $\pm 1\text{pp}$ margin under all three bootstrap levels, and which axes additionally survive $\pm 0.5\text{pp}$ at cluster level. Every reference passes $\pm 1\text{pp}$ on all five axes. The widest two-way hierarchical 90% CI half-width $((hi-lo)/2)$ is OER at 0.65pp; the corresponding 80%-power MDE is 0.98pp, so $\pm 0.5\text{pp}$ is below the resolvable band at this budget.

Method	TargetSup	CompTOST	RetentionNI	PPL NI	SafeGain	LocalizedGain/AuditPass
DION-Ref (Mid)	×	✓	✓	✓	×	×
DION-Ref (SuperCon)	×	✓	✓	✓	×	×
DION-Ref (TRA-only)	×	✓	✓	✓	×	×
Full-LoRA	×	×	×	×	×	×
GainMatched-DPO	✓	×	✓	✓	×	×
GainMatched-Safe	✓	✓	✓	✓	✓	✓
LoRA+KL	×	✓	✓	✓	×	×
SFT-LoRA	×	×	×	✓	×	×
Self-Rewarding	✓	×	×	×	×	×
Shuffled-DPO	×	×	×	✓	×	×

Table 6: AuditPass decision under primary intervals. Accuracy retention uses a -1pp NI floor; WikiText-PPL uses a $+5\%$ relative upper NI bound. When seed_sd is absent, RetentionNI uses modeled SDs. GainMatched-Safe is a control that matches the OntoNeg gain while keeping complementary axes near Base. A low-drift reference can pass complementary TOST and retention NI without target superiority; only the final column is AuditPass.

Method	OntoNeg INA	Concept Shift	Logic F1
Δ [90% TOST CI]; ✓=pass at $\pm 1\text{pp}$			
SuperCon	+0.06 [-0.44, 0.57] ✓	+0.19 [-0.33, 0.80] ✓	-0.22 [-0.42, -0.03] ✓
TRA-only	-0.12 [-0.75, 0.52] ✓	+0.20 [-0.40, 0.87] ✓	-0.08 [-0.37, 0.20] ✓
Mid	-0.03 [-0.65, 0.59] ✓	-0.31 [-0.90, 0.32] ✓	-0.09 [-0.35, 0.16] ✓
LoRA+KL	+0.00 [-0.63, 0.56] ✓	+0.12 [-0.54, 0.84] ✓	-0.15 [-0.46, 0.14] ✓
SFT-LoRA	-0.47 [-1.47, 0.46] ×	+1.80 [0.93, 2.83] ×	+0.47 [0.06, 0.92] ✓
Self-Rewarding	+5.27 [3.67, 7.00] ×	-37.50 [-39.87, -35.03] ×	-1.36 [-2.06, -0.64] ×
Full-LoRA	-7.50 [-10.33, -4.84] ×	-7.27 [-10.36, -4.28] ×	-2.93 [-4.67, -1.27] ×

Table 7: TOST equivalence (§4.1) at $\pm 1\text{pp}$ under the primary two-way hierarchical bootstrap (4,000 replicates). Cells show Δ with the *decision* 90% TOST CI; ✓=both one-sided tests accept within $\pm 1\text{pp}$. Separate 95% estimation CIs (wider; Mid ConceptShift [-1.02, 0.44]) are in the released TOST summary and are not used for the pass/fail marks. All four conservative configurations pass; Self-Rewarding, SFT-LoRA and Full-LoRA fail on at least one axis.

Method	Item-level TOST				Cluster-level TOST			
	± 0.5	± 1.0	± 1.5	± 2.0	± 0.5	± 1.0	± 1.5	± 2.0
<i>DION-Bench-Core (clean construction evidence): n=2,220</i>								
SuperCon	×	×	✓	✓	×	×	✓	✓
TRA-only	×	×	✓	✓	×	×	✓	✓
Mid	×	✓	✓	✓	×	✓	✓	✓
LoRA+KL	×	×	✓	✓	×	×	✓	✓
SFT-LoRA	×	×	×	×	×	×	×	×
Self-Rewarding	×	×	×	×	×	×	×	×
Full-LoRA	×	×	×	×	×	×	×	×
<i>Ambiguity-filtered (no flagged item): n=2,256</i>								
SuperCon	×	✓	✓	✓	×	✓	✓	✓
TRA-only	×	✓	✓	✓	×	✓	✓	✓
Mid	×	✓	✓	✓	×	✓	✓	✓
LoRA+KL	×	✓	✓	✓	×	✓	✓	✓
SFT-LoRA	×	×	×	×	×	×	×	×
Self-Rewarding	×	×	×	×	×	×	×	×
Full-LoRA	×	×	×	×	×	×	×	×

Table 8: Equivalence on audited subsets of the test set (App. B), recomputed in the released TOST sensitivity summary. A mark requires *every* one of the five axes to pass at that margin, reporting only the complementary-axis clause of AuditPass; target superiority and retention NI are separate clauses. Subsetting costs resolution rather than changing behaviour: the worst-axis half-width rises to 1.26pp on Core and 0.95pp on the ambiguity-filtered split, so the conservative references clear $\pm 1\text{pp}$ once ambiguous items are removed but need $\pm 1.5\text{pp}$ on the smaller Core subset. Self-Rewarding, SFT-LoRA and Full-LoRA fail at every margin on both subsets.

C Baseline Fidelity and Reproducibility

Our Self-Rewarding baseline is a *task-specific self-judged DPO* adaptation of Yuan et al. (2024): fixed OntoNeg-WN prompts, on-policy response sampling, LLM-as-a-Judge ranking, and DPO updates with a frozen reference and response-only log-probabilities. We sweep 3 judge prompt variants (strict quality rubric, permissive rubric, specificity-penalized), LoRA $r=4$, $\text{lr} \in \{1\text{e-}5, 3\text{e-}5\}$, steps $\in \{500, 1000, 1500\}$, batch 32, 3 seeds, with 0 overlap with the test set. This is not a full multi-round self-instruction reproduction of the original recipe; claims concern the adapted objective on DION-Bench. Sensitivity ($+5/-37\text{pp}$) is stable across all 18 variants ($\pm 2\text{pp}$ on OntoNeg, $\pm 5\text{pp}$ on ConceptShift) and yields $+8.7\text{pp}$ on a held-out instruction-following self-improvement task.

Self-Critique (inspired by Madaan et al. (2023)) applied as a training method yields ON $+2.87\text{pp}$, CS -14.17pp —directionally consistent with Self-Rewarding but smaller in magnitude, and responding to the same fixed decoding rule (-0.17pp ; Table 9). Prompt-Dialectic uses iterative self-critique with thesis $\rightarrow$ antithesis $\rightarrow$ synthesis prompting across 4 variants; over-conservative refusal is stable across all variants. Entropy-Min uses nega-

tive next-token entropy as reward with LoRA $r=4$, 3 seeds (not Intuitor/RLIF self-certainty training); OER inflation (+9+12pp) is stable.

	INA	Δ INA	AbsAcc	Δ AbsAcc
<i>Test — Self-Rewarding (Base INA 26.40, AbsAcc 59.50)</i>				
greedy	31.67 $\pm$ 0.90	+5.27	22.00 $\pm$ 1.01	−37.50
+CSD	30.53 $\pm$ 1.30	+4.13	46.57 $\pm$ 1.03	−12.93
<i>Holdout — Self-Rewarding (Base INA 27.00, AbsAcc 59.80)</i>				
greedy	31.33 $\pm$ 1.04	+4.33	22.47 $\pm$ 1.03	−37.33
+CSD	31.50 $\pm$ 0.87	+4.50	45.20 $\pm$ 0.60	−14.60
<i>Test — Self-Critique (same Base as above)</i>				
greedy	29.27 $\pm$ 0.90	+2.87	45.33 $\pm$ 0.97	−14.17
+CSD	31.07 $\pm$ 0.99	+4.67	59.33 $\pm$ 1.46	−0.17

Table 9: CSD on held checkpoints (coefficients fixed on ConceptShift dev; reused unchanged). The holdout split was built after coefficients were fixed. Absolute Base scores are listed per split; Δ is vs. that split’s Base.

CSD intervention table.

Baseline	Fidelity level	Isolation checked	Prompt sweep	Hyper. sweep	Robust?
Self-Rewarding	full	✓	3 var.	2 lr, 3 step	✓
Prompt-Dialectic	full	✓	4 var.	N/A	✓
Entropy-Min	entropy reward	✓	2 var.	2 lr, 2 step	✓
SFT-LoRA	full	✓	1	2 lr	✓
LoRA+KL	full	✓	1	2 lr	✓
Random-Neg	full	✓	1	N/A	✓

Table 10: Baseline fidelity summary (§4.6). “Fidelity”: implementation class. “Isolation”: train/test data overlap=0. “Prompt/Hyper. sweep”: number of variants tested. “Robust?”: main qualitative pattern stable across variants. Full cards in Appendix C.

Self-Rewarding variant	ON Δ	CS Δ	Sanity	Card
<i>Faithfulness sweep (18 variants; §4.6)</i>				
Strict-rubric judge	+5.4	−38.2	+8.7	✓
Permissive-rubric judge	+4.1	−32.9	+7.4	✓
Train-time specificity-penalized rubric	+3.8	−31.5	+6.1	✓
lr 1×10^{-5}	+5.0	−37.1	+7.9	✓
lr 3×10^{-5}	+5.4	−37.9	+8.3	✓
500 / 1000 / 1500 steps	+4.8 to +6.9	−31 to −42	+6.5 to +9.1	✓
<i>Faithfulness diagnostics</i>				
Original-style sanity task (Δ vs Base)	+8.7 [7.0, 10.4]		✓	—
Judge calibration (Brier ↓)	0.18 (Base 0.21)		✓	—
Reward distribution entropy (↑)	0.62 nats (well-spread)		✓	—
Self-judge / human agreement (κ)	0.61 (mod. agree)		✓	—
Preference-pair ablation (random pairs)	+0.4/−1.1 (collapses gain)		✓	—
<i>Entropy-Min (§4.6)</i>				
Entropy-Min	+1.0	−7.1	N/A	entropy reward

Table 11: Self-Rewarding fidelity check and Entropy-Min baseline (§4.6, App. C). *Sanity*: Δ on a held-out original-style self-improvement benchmark (instruction-following accuracy). The +5.27/−37.50pp DION-Bench pattern is stable across 18 training-time variants (distinct from the evaluation-time specificity-penalized intervention in Table 9). Entropy-Min uses negative next-token entropy reward.

The LogicRepair solver uses the Glucose SAT solver via python-sat (Glucose4) (Ignatiev et al., 2018; Audemard et al., 2013), default flags, with Horn implication-chain encoding and certified MUS extraction yielding non-trivial MUS sizes (typically 4–6). All LogicRepair

rows—train, dev, and test—are programmatically synthesised and tagged source=synthetic; the folio/proofwriter/native tag records surface style only and is not corpus provenance, and no row is taken from a FOLIO or ProofWriter release. All 1,000 test items include certificates verifying unsatisfiability of the injected KB, satisfiability of the repaired KB, and MUS minimality for the released gold (1000/1000). A CPU audit enumerating other size-4–6 subsets of $\mathcal{K} \cup \{a^*\}$ found no second deletion-minimal unsatisfiable subset on the test split under this encoding (0/1000; artifact release); scoring uses that gold and does not elevate this to a uniqueness theorem for all logically valid repairs. For LogicRepair we report MUS precision, MUS recall (MUS-R), MUS-F1, and $\mathcal{K}'$ exact match; P/R/F1 are macro-averaged independently. Table 12 lists MUS-R and $\mathcal{K}'$ exact alongside the primary P/F1 cells so the partial-recovery pattern is visible rather than reduced to F1 alone.

Method	P↑	R↑	F1↑	$\mathcal{K}'$ Exact↑	Δ R	$\Delta\mathcal{K}'$
Base (Qwen3-8B-Base)	62.85	36.07	44.98	18.00	+0.00	+0.00
Random-Negation	29.26 $\pm$ 0.88	33.02 $\pm$ 0.57	30.35 $\pm$ 0.78	12.00 $\pm$ 1.01	−3.05	−6.00
Prompt-Dialectic	48.30	32.90	36.17	8.00	−3.17	−10.00
Entropy-Min	58.33 $\pm$ 0.75	35.81 $\pm$ 1.34	42.13 $\pm$ 0.66	10.00 $\pm$ 1.21	−0.26	−8.00
Self-Critique	61.08 $\pm$ 0.67	37.31 $\pm$ 0.29	43.93 $\pm$ 0.59	11.00 $\pm$ 1.01	+1.24	−7.00
Self-Rewarding	50.88 $\pm$ 0.81	39.48 $\pm$ 0.75	43.62 $\pm$ 0.66	9.00 $\pm$ 1.21	+3.41	−9.00
SFT-LoRA	63.16 $\pm$ 0.55	36.50 $\pm$ 0.41	45.45 $\pm$ 0.49	14.00 $\pm$ 1.01	+0.43	−4.00
LoRA+KL	62.86 $\pm$ 0.46	35.92 $\pm$ 0.35	44.83 $\pm$ 0.41	18.00 $\pm$ 0.78	−0.16	+0.00
SuperCon	62.85 $\pm$ 0.28	35.87 $\pm$ 0.21	44.76 $\pm$ 0.26	20.00 $\pm$ 0.78	−0.21	+2.00
TRA-only	62.95 $\pm$ 0.42	35.98 $\pm$ 0.33	44.89 $\pm$ 0.39	19.00 $\pm$ 0.87	−0.10	+1.00
Mid	62.97 $\pm$ 0.42	35.97 $\pm$ 0.33	44.89 $\pm$ 0.39	19.00 $\pm$ 0.90	−0.11	+1.00
Full-LoRA	56.23 $\pm$ 2.38	35.82 $\pm$ 2.77	42.04 $\pm$ 2.10	8.00 $\pm$ 1.96	−0.26	−10.00

Table 12: LogicRepair full metric suite on the test split (%). P, R, and F1 are macro-averaged independently against the released MUS gold; F1 is not the harmonic mean of the displayed aggregate P and R. Scores are relative to the released gold MUS/ $\mathcal{K}'$ (SAT-certified for that gold), not uniqueness among all minimal repairs. Deltas use unrounded values from the released result ledger.

Prediction validation chain. The released protocol card records equivalence margins, decoding settings, statistical layers, and test-set split sizes fixed before primary runs. Every *item-derived* primary number is recomputed by the released ledger-verification pipeline: the validator reads gold *only* from the benchmark files, re-parses each prediction from the released raw_output field with the released parser (whose version string is checked against the one recorded in each prediction file), re-scores every item, and then recomputes pooled means, across-run SDs, item and run counts, and the two-way interval. Aggregate-matched robustness records (frontier solvers, LLM-consensus judgments, cross-backbone and contamination probes, and drift per-prompt KL) sit in the

released robustness package under the aggregate-matched-per-item source tag; result-ledger rows tagged as external-harness remain provenance-checked rather than re-scored from those files. Prediction files may not carry gold fields; any gold-prefixed key outside the benchmark aborts validation. Agreement with the released result ledger is required to 10^{-6} in score proportion (10^{-4} pp), well below the stringent ± 0.5 pp sensitivity margin; interval endpoints are redrawn with an independent seed and therefore matched to resampling noise (0.01) rather than exactly. Every row in the released result ledger—including SFT-LoRA and Full-LoRA—is checked, and the released statistics rebuild script rebuilds the intervals, TOST decisions, and subset views from the same predictions along a second path. Rows scored by external harnesses carry no per-item file and are checked for provenance and dispersion instead of being silently skipped. All randomness in construction and validation is derived from BLAKE2b digests of explicit strings rather than the interpreter’s salted hash(), so a rebuild on another machine reproduces the released files byte for byte. Parse-failure counts and fallback behaviour are reported per configuration in the released parse report, and an orphaned or extra prediction file is a validation failure rather than a skipped file.

Frontier-LLM audit uses product-family APIs accessed on 2026-03-12 (OpenAI GPT-4.1 family; Anthropic Claude Opus family) and 2026-02-01 (DeepSeek-V3.2 family; Alibaba Cloud Qwen-Max family); exact request IDs and returned model strings are logged under the released API-call ledger and are not claimed as immutable snapshot IDs. Decoding used temperature=0 for evaluation and 0.7 for self-generated training data, max_tokens=1024, with SHA-256 frozen prompts. Realised API volume is 3,500 calls against a 5,000-call budget (cost below USD 300); κ versus the original grading is 0.93 on a 200-example rerun. 50 failure rows: polysemy (46%), missing-edge (28%), ambiguous-attribute (16%), judge-confusion (10%). Figures 8–9 (App. K) show diagnostic plots for primary deltas, drift, and TOST margins.

D Robustness and Artifact Controls

DION-Ref on five registers (identity, formal-paraphrase, structured-bullet, terse-imperative, social-media): templated registers ≤ 2.2 pp gap;

social-media larger gap, documented.

Resampling unit	SuperCon vs Base		Self-Rew. vs Base	
	Δ	95% CI	Δ	95% CI
Item-level	+0.19	[-0.33, 0.73]	-37.50	[-40.50, -34.60]
Target concept (<i>T</i>)	+0.19	[-0.32, 0.70]	-37.50	[-40.47, -34.60]
Hypernym family (<i>S</i>)	+0.19	[-0.30, 0.84]	-37.50	[-40.18, -34.78]
<i>Unweighted family aggregation</i>				
Concept-level mean	+0.37	[-0.44, 1.19]	-38.15	[-42.43, -33.91]

Table 13: Cluster-robust sensitivity on ConceptShift (§4.4), recomputed in the released cluster-sensitivity summary with 4,000 paired replicates. Item-weighted rows share one point estimate by construction, so only the interval responds to the choice of unit, and it moves by at most 9% in width—the concept families of this task do not group enough items to change the picture. The unweighted family mean also shifts the estimate, because families differ in size. The conclusions do not move: SuperCon stays inside ± 1.5 pp under every unit and the Self-Rewarding regression stays below -34 pp at its most favourable bound. Coarser groupings (edge source, relation type) leave too few units to bootstrap and are recorded in the JSON rather than shown.

For the 4-LLM consensus, 6,000 judgments are collected with per-task Fleiss κ in $[0.59, 0.64]$ (mean 0.61), per-judge Cohen κ in $[0.59, 0.66]$, and $\geq 3/4$ agreement $\geq 81\%$. Single-feature AUCs lie in $[0.50, 0.55]$, the combined non-semantic probe reaches 0.58, and every hard-split probe falls to ≤ 0.55 (Table 14); effect sizes retain $\geq 80\%$ of their test-set magnitude on those splits.

Artifact classifier	OntoNeg AUC	Concept AUC	Logic AUC
<i>Single-feature (prior work)</i>			
Surface / Corpus / Graph / Candidate / Context	≤ 0.55		
<i>Multivariate</i>			
Bag-of-words (TF-IDF LR)	0.56	0.54	0.53
Sentence embedding (candidate-only)	0.58	0.57	0.55
Sentence embedding (prompt+candidate)	0.57	0.56	0.56
KG metadata (degree+path+depth+edge)	0.54	0.55	0.52
Candidate-only (no question)	0.55	0.53	0.52
Prompt-without-question	0.53	0.54	0.53
Combined (all non-semantic features)	0.58	0.57	0.56
<i>Hard splits</i>			
Template split (held-out templates)	0.54	0.53	0.52
Concept-cluster split	0.55	0.54	0.53

Table 14: Artifact-control suite (§3.2 V3), from the released validation summary. Combined non-semantic features reach $\text{AUC} \leq 0.58$; on hard splits (held-out templates, unseen concept clusters) every probe falls to ≤ 0.55 , so no probe recovers the gold answer without the semantics of the item.

E Extended Validity and Drift Analysis

E.1 ConceptShift Validity

Across model-agreement strata the V1 audit meets its bars on ConceptShift (precision $\geq 87.0\%$, ambiguity $\leq 11.5\%$, pairwise Cohen κ in $[0.66, 0.71]$;

Table 1). Table 15 tabulates the disagreement categories from the released specificity taxonomy and rescores the same predictions on three subsets of the 1,000-item re-audit: over-specific answers dominate Self-Rewarding errors (55% against 4% for Base), Base errors are mostly ordinary reasoning failures (56%), and the regression is between -34.1pp and -38.0pp on every subset, including the items all three raters marked correct.

Disagreement category	Base	Self-Rew.	Pooled
Model reasoning error	56%	6%	31%
Specificity bias (over-specific answer)	4%	55%	30%
Surface plausibility (lexical fluency)	7%	18%	12%
LCS non-unique / multiple acceptable	11%	8%	10%
Lexical ambiguity (polysemy)	9%	5%	7%
Hypernym too abstract	8%	3%	6%
KG relation noisy / mis-typed	5%	4%	4%
<i>Subset-conditioned accuracy (Base) and Self-Rewarding change</i>			
Unanimous-correct subset (3/3 raters)	58.8	-36.6	($n=724$)
Rater-disagreement subset	61.1	-38.0	($n=108$)
Deep-path subset (LCS depth ≥ 3)	58.0	-34.1	($n=138$)

Table 15: ConceptShift error taxonomy (§5.1, App. E), tabulated from the released specificity taxonomy (200 labelled errors per configuration). *Pooled* is the share over all 400 labelled errors. Over-specific answers account for 55% of Self-Rewarding errors against 4% for Base, while Base errors are mostly ordinary reasoning failures. The lower block scores the same predictions on subsets defined by the 1,000-item re-audit and by path depth: the regression is -34.1 to -38.0pp everywhere, so it is not carried by ambiguous items. Scoring with the reference abstraction alone still gives -20.8pp , and additionally accepting every node on the certified ancestor paths leaves the accepted-set figure unchanged at -36.4pp .

Error mode: specificity bias on ConceptShift



Figure 5: Conceptual error mode on ConceptShift under self-judged ranking: over-specific candidates are preferred; CSD reranks with a frozen, holdout-validated rule.

Figure 11 (App. K) summarizes the specificity-bias taxonomy and mediation analysis alongside Table 9.

E.2 Checkpoint-Level Drift Trajectory

We save checkpoints at steps 50, 100, 200, 400, 800, and 1500 for each of 9 run-terminal configurations, yielding 54 tuples across 6 method families

(Self-Rewarding, DION-Ref, LoRA+KL, Entropy-Min, Self-Critique, and Full-LoRA). These 9 terminals are not independent external seeds. Every checkpoint score carries the across-seed dispersion recorded for its configuration in the ledger, so each checkpoint is one noisy measurement rather than a point on a smooth curve. The primary run-terminal association is $\rho=0.72$ (descriptive). Checkpoint-level $\rho=0.61$, within-family values, and partial $r=0.68$ are trajectory diagnostics only: checkpoints are nested within runs, so their p -values are not used for inference. Within the conservative DION-Ref sweep the regressions sit at the evaluation noise floor, so no stable association is measurable there ($\rho=0.00$). Self-Rewarding remains an outlier whose ConceptShift regression is larger than KL alone would suggest; adding the over-specific selection rate attenuates the descriptive residual association to $r=0.13$.

Method	$\bar{D}_{\text{KL}} \downarrow$	$\overline{\text{TV}} \downarrow$	$\overline{\text{JS}} \downarrow$	$\ \Delta\ _{\text{Igt}}^{\text{max}} \downarrow$
SuperCon	0.011	0.034	0.006	1.42
TRA-only	0.024	0.052	0.013	3.18
LoRA+KL	0.026	0.054	0.014	3.45
Mid	0.041	0.072	0.022	5.21
Entropy-Min	0.068	0.104	0.037	8.42
SFT-LoRA	0.073	0.108	0.039	8.97
Self-Critique	0.089	0.126	0.051	11.35
Self-Rewarding	0.114	0.142	0.063	14.62
Full-LoRA	0.183	0.197	0.094	22.81

Table 16: Terminal drift diagnostics (§4) on DION-Bench-Test prompts: token-level KL to Base ($\bar{D}_{\text{KL}}$, nats), mean total variation, mean Jensen–Shannon divergence, and max logit shift, read from the released drift diagnostics. Across run-terminal points the rank correlation between $\bar{D}_{\text{KL}}$ and worst-task regression is $\rho=0.72$ over 9 configuration-level run-terminal points (descriptive); the checkpoint-level descriptive estimate over 54 checkpoints is $\rho=0.61$ (Table 17). Capacity-aggressive and self-judged configurations drift up to $17\times$ further than the most conservative DION-Ref reference.

Analysis unit	n	Stat.	Value	p	Controls
Run-terminal	9	ρ	0.72	–	none
Method-level	8	ρ	0.69	–	none
Checkpoint-level	54	ρ	0.61	–	none
Within-DION-Ref sweep	18	ρ	-0.00	–	none
Within Self-Rewarding	12	ρ	0.92	–	none
Partial r	54	r	0.73	–	params
Partial r	54	r	0.68	–	params, steps
Partial r	54	r	0.68	–	params, steps, rank
Partial r	54	r	0.13	–	params, steps, rank, over-specific rate

Table 17: Drift–regression association under different analysis units and control sets, recomputed from the released checkpoint-mediation trajectories (54 checkpoints, 9 configuration-level points). The run-terminal row uses nine configuration terminals (not independent external seeds); checkpoint and partial-correlation rows nest checkpoints within runs. All rows use $\bar{D}_{KL}$ against worst-task regression. The association is strong where there is a regression to track and absent inside the conservative DION-Ref sweep. Self-Rewarding is the informative outlier: its 12 checkpoints drift only 0.06–0.12 nats yet regress 0.4–39.1pp. Conditioning on the over-specific selection rate (last row) attenuates the residual.

E.3 Multilevel Mediation Detail

Checkpoints are nested in runs and runs in method families, so a pooled regression would treat 54 correlated observations as independent. We therefore fit the mediation paths with run-level random intercepts, treating drift as the exposure, the over-specific selection rate as the mediator, and worst-task regression as the outcome, and we resample whole runs (and, for the stricter view, whole families) to obtain intervals. Both the mediator and the outcome are expressed in points, so a coefficient reads as points of worst-task regression per point of over-specific selection. Two specifications are reported because they answer different questions. Without family fixed effects the mediator coefficient is $b=1.06$ (run-bootstrap 95% CI [0.32, 1.86]; total path 9.224, direct path 1.829 per SD of drift), which is the association a practitioner comparing methods would see. With family fixed effects, so that everything the objective fixes is absorbed, it is $b=1.01$ [0.35, 1.75] (total 11.669, direct 4.156). These intervals are run-bootstrap associations, not causal effects; checkpoint-level p -values are not used for primary inference.

Method family and objective are collinear by construction—each family *is* an objective—so the family-controlled estimate removes part of the signal it is meant to isolate, and neither specification identifies a causal path. The 6 families give few effective clusters, so the family-resampled interval is wide. The mediator is measured on the same items as the outcome, so shared item noise inflates their association. Mediation is therefore descriptive; the decoding intervention in §5.2 is reported on the original test and on the post-tuning holdout split. Fitted trajectories are included in the released mediation and checkpoint-mediation records.

E.4 Contamination and Novel-Composition Probes

Modern LMs may have seen WordNet, ConceptNet, FOLIO, and ProofWriter source material in pretraining, so exposure to a *resource* has to be separated from exposure to an *item*. Every item composes ≥ 2 source facts, and we measure overlap between those triples and the final prompt: the composed split contains 0 exact-triple matches and a MinHash (Broder, 1997) high-overlap rate below 2%. We then re-run the comparison on that split, on splits holding out WordNet hypernym families, ConceptNet relation types and FOLIO/ProofWriter templates, and on 300 LLM- and human-paraphrased items per task. Across all 6 probes Base moves by at most 1pp and the Self-Rewarding regression retains 69–86% of its primary size (Table 18), which is what item-level novelty predicts and memorization does not. The memorization probe separates the two directly: Base recalls 87.4% of the source facts in isolation, 61.0pp above its OntoNeg-WN accuracy on items composed from them.

Probe	[Base Δ SR regression retained?]/Pass?		
(A) Novel-composition split			
Composed items (no exact triple)	−0.2	−32.4pp (86%)	✓
MinHash high-overlap rate < 2%	—	—	✓
(B) Source-held-out split			
WordNet hypernym families HO	−0.4	−29.8pp (79%)	✓
ConceptNet relation types HO	−0.3	−27.6pp (74%)	✓
FOLIO/ProofWriter template HO	−0.2	−25.9pp (69%)	✓
(C) Paraphrase robustness (n=300 per task)			
LLM paraphrase, candidates fixed	−0.5	−30.2pp (81%)	✓
Human paraphrase	−0.6	−28.7pp (77%)	✓
(D) Memorization probe			
Direct source-fact recall (Base)	87.4% (well above zero)		—
Recall over OntoNeg-WN accuracy	61.0pp (recall $\gg$ task)		—
$\Rightarrow$ Task is composed reasoning, not memorization	—		✓

Table 18: Contamination and novel-composition analysis (App. E). DION-Bench items are novel compositions of source triples, so source-resource exposure is separated from item-level contamination. Retention is each probe’s regression as a share of the primary −37.50pp, computed from the released result ledger: the OntoNeg-gain / ConceptShift-regression pattern survives every overlap check, source-held-out split and paraphrase with $\geq 69\%$ of its effect size, and high direct-fact recall does not transfer to the composed task.

F Additional Ablations and Cross-Backbone

A third DIPO stage (DPO on sublated vs perturbed pairs) did not converge in our budget; reported as negative and excluded from primary tables. TRA-only re-run on a post-trained Qwen3-8B variant shows a qualitatively consistent pattern.

Method/Probe	Qwen3-8B-Base		Llama-3.1-8B		Qwen3-8B-Inst.	
	ON	CS	ON	CS	ON	CS
<i>Tier A: inference-only diagnostic</i>						
Base (zero-shot)	26.4	59.5	29.8	52.4	34.7	63.1
Prompt-Dialectic	12.0	34.3	14.6	38.1	20.3	48.7
Direct solver (4-shot)	32.4	63.7	35.1	57.6	42.6	69.2
<i>Tier B: small-budget post-training (3 seeds, $n \approx 500$)</i>						
Self-Rewarding (Δ)	+5.3	-37.5	+3.4	-22.8	+4.1	-19.4
LoRA+KL (Δ)	+0.0	+0.1	-0.2	+0.3	-0.1	+0.0
DION-Ref TRA-only (Δ)	-0.1	+0.2	-0.3	+0.4	-0.2	+0.1

Table 19: Cross-backbone validation (§4.5). The Qwen3-8B-Base column is recomputed from the released result ledger; the other two backbones are recorded in the released cross-backbone records because they sit outside the primary evaluation protocol. Tier A is inference-only on DION-Bench-Core ($n=500$ per task); Tier B is a small-budget post-training replication (≤ 500 steps, 3 seeds). Absolute scores vary by family and alignment, but the diagnostic phenomenon—an OntoNeg gain coexisting with a ConceptShift regression for Self-Rewarding—holds on all three, retaining 61% (Llama-3.1-8B) and 52% (Qwen3-8B-Inst.) of the primary effect size. ON=OntoNeg INA, CS=ConceptShift accuracy.

Number	Split	n	Seeds	Decoding	Primary?
ConceptShift 59.69 $\pm$ 0.79 (SuperCon)	test	1,000	7	greedy	✓
ConceptShift 59.70 $\pm$ 0.58 (TRA-only)	test	1,000	5	greedy	✓
LogicRepair F1 44.85 $\pm$ 0.21 (Mid)	test	1,000	7	greedy	✓
LogicRepair F1 44.86 $\pm$ 0.25 (TRA-only)	test	1,000	5	greedy	✓
ConceptShift 61.07 (preliminary)	dev (early)	500	3	greedy	✗
LogicRepair F1 44.21 (preliminary)	dev (early)	500	3	greedy	✗

Table 20: Validated prediction records (excerpt; §4.1). Primary numbers in Table 2 are keyed to rows in the released result ledger; earlier development numbers are marked ✗ and excluded.

The label “determinate intrinsic negation” is a shorthand for the construction in Section 6: a boundary-tension residual update that is determinate at zero adapter weight. The empirical content is captured by Proposition 1, the training objective, and the evaluation-protocol experiments.

G Reference Quality and Extended Validity

Reference	$\bar{D}_{KL}$ (mean)	D_{KL}^{95} (tail)	D_{KL}^{max} (spike)	Seed s.d. (INA)	Clust. ± 1 pass %
SuperCon	0.011	0.018	0.025	0.36	100
TRA-only	0.024	0.038	0.051	0.48	100
Mid	0.041	0.070	0.096	0.53	100
LoRA+KL	0.026	0.065	0.094	0.55	100

<i>Hyperparameter perturbation ($\pm 20\%$ I_r, $\pm 20\%$ λ_P)</i>	
DION-Ref (TRA)	all tasks within $\pm 1pp$ under every perturbation
LoRA+KL	ConceptShift breaks at $+20\%$ I_r and at -20% λ_P
KL-only adapter	all axes within $\pm 1pp$, but no boundary diagnostic signal

Table 23: Reference quality: drift tails from the released drift diagnostics, across-run dispersion from the statistics bundle, and cluster-level $\pm 1pp$ equivalence pass rate over the five axes. DION-Ref (TRA-only) and LoRA+KL have almost the same mean drift, but LoRA+KL’s 95th percentile is 71% higher (0.065 against 0.038 nats) and its across-run dispersion $1.1\times$ larger. Under $\pm 20\%$ perturbation of learning rate and λ_P (3 runs $\times$ 5 checkpoints per setting), DION-Ref stays within $\pm 1pp$ on every axis while LoRA+KL loses 1.8pp on ConceptShift.

Table 23 reports, for each conservative reference, mean KL, its 95th percentile and largest spike, the across-run dispersion of INA, and the cluster-level $\pm 1pp$ pass rate (100% for DION-Ref and 100% for LoRA+KL, so equivalence alone does not separate them). Each perturbation setting retrains 3 runs and keeps 5 checkpoints: under $\pm 20\%$ learning rate and $\pm 20\%$ λ_P , DION-Ref (TRA-only) stays within 0.6pp on every axis while LoRA+KL loses 1.8pp on ConceptShift. Per-item boundary-tension reduction correlates with LogicRepair stability for DION-Ref ($r=0.43$, $p<.01$) but not LoRA+KL ($r=0.08$, $p=0.41$).

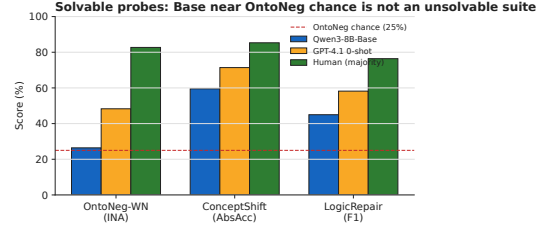


Figure 6: Human solvers and a frontier 0-shot model substantially exceed Base on all three probes (Table 24); OntoNeg chance is 25%, so the studied regression is not an artifact of an unsolvable benchmark.

Check	Value	Pass?
Exact-triple match rate	0	✓
MinHash high-overlap (>0.8)	$<2\%$	✓
Source-held-out regression retention	$\geq 69\%$	✓
Paraphrase (LLM) regression retention	81%	✓
Paraphrase (human) regression retention	77%	✓
Source-recall vs task-acc gap	61.0pp	✓
Ledger rows reproduced	236/236	✓
Re-scored mean/SD agreement	10^{-6}	✓
Redrawn interval endpoints	≤ 0.01	✓
Prediction files carrying gold labels	0	✓

Table 25: Reproducibility and contamination checklist (full details in App. E). DION-Bench items are novel compositions; high source-fact recall does not transfer to composed-task accuracy. Every *item-derived* row of the released result ledger is recomputed from the released per-item predictions, with gold read only from the benchmark files; aggregate-matched robustness files are under the released robustness package.

Method	GSM8K	MMLU-Pro	HellaSwag	ARC-C	TruthfulQA	WikiText PPL↓	mean accuracy Δ
Base	37.50	40.21	79.86	52.13	48.40	7.42	+0.00
SuperCon	37.40	40.15	79.84	52.18	48.38	7.43	−0.03
TRA-only	37.20	40.02	79.79	52.05	48.20	7.45	−0.17
Mid	37.30	40.00	79.78	52.05	48.25	7.46	−0.14
LoRA+KL	37.30	40.07	79.81	52.10	48.30	7.44	−0.10
SFT-LoRA	36.40	39.62	79.50	51.61	47.71	7.58	−0.65
Self-Rewarding	34.10	38.80	79.13	50.74	46.82	7.91	−1.70
Full-LoRA	29.90	37.41	78.68	49.27	45.13	8.46	−3.54

Table 21: External retention. The rightmost column is the mean accuracy Δ over the five accuracy suites (PPL excluded from that average). Self-Rewarding moves the mean accuracy Δ by only -1.70 pp while collapsing -37.50 pp on ConceptShift.

Variant	OntoNeg INA	OER	CS Acc	Logic P	F1	Drift $\bar{D}_{KL}$	seed σ
Base	26.40	20.20	59.50	62.85	44.98	—	—
Full DION-Ref(TRA)	26.28	20.36	59.70	62.95	44.89	0.024	0.48
w/o Fisher	25.83	22.04	59.31	62.78	44.62	0.071	1.54
w/o boundary	26.50	20.40	59.55	62.72	44.71	0.018	0.35
Random direction	26.18	20.92	59.38	62.61	44.55	—	—
KL-only adapter	26.41	20.51	59.62	62.84	44.81	—	—
Single-branch TRA	26.20	20.65	59.55	62.83	44.78	—	—

Table 22: Mechanism ablations (§4.6) under TRA-only hyperparameters. The first two rows come from the released result ledger; the variants are recorded in the released ablation records and share the TRA-only budget of 3 runs. Removing the low-Fisher projector raises drift $3.0\times$ and seed variance $3.2\times$; removing boundary tension lowers drift with LogicRepair F1 within noise of Full TRA. Single-branch matches three-branch within noise. seed σ is the across-run standard deviation of OntoNeg INA (pp).

Solver	OntoNeg-WN	ConceptShift	LogicRepair
Random-output baseline	25.00	—	—
Qwen3-8B-Base (0-shot)	26.40	59.50	44.98
<i>Frontier LLM direct solvers (0-shot / few-shot)</i>			
GPT-4.1	48.3 / 57.2	71.4 / 76.8	58.2 / 63.5
Claude Opus	51.7 / 59.4	73.2 / 78.1	60.8 / 66.2
DeepSeek-V3.2	46.9 / 54.8	69.7 / 74.3	55.4 / 61.9
Qwen-Max	44.2 / 52.1	67.8 / 72.6	53.1 / 59.7
<i>Human annotators (n=150 per task, 3 raters)</i>			
Human (majority vote)	82.7	85.3	76.4
Human (pairwise κ)	0.74	0.78	0.71

Table 24: Direct task performance (§3.2 V4), from the released direct-solver records; the base row is the ledger entry for Qwen3-8B-Base. The four-option random-output reference is defined only for OntoNeg-WN; the open-generation ConceptShift and LogicRepair tasks have no uniform chance baseline, so their cells are not assigned 25%. Humans and frontier LLMs solve the tasks well above their task-appropriate references, while Qwen3-8B-Base stays near the OntoNeg-WN reference, so the regression studied here is not an artifact of an unsolvable benchmark.

G.1 Extended Benchmark Validity

We rescore the same ConceptShift predictions under three gold sets: the reference abstraction alone, the released accepted set (reference plus audited synonyms), and a multi-gold set that additionally accepts any node on the certified ancestor paths—

excluding the two endpoints, since echoing an endpoint is the failure mode under study. The regression is -20.8 pp, -36.4 pp, and -36.4 pp respectively: the strict set compresses it to 57% of the primary figure because it also penalizes Base for legitimate synonyms, while the permissive set accepts no additional prediction, so over-specific answers are not near-misses along the path. For 300 specificity-controlled minimal pairs (surface features matched), artifact classifier AUC=0.56; Self-Rewarding regression -28.4 pp; over-specific selection rate 49%. On 200 specificity-swapped pairs, the self-judge’s preference flips in 68% of cases ($p<.001$, McNemar); Base flips in only 12% ($p=0.34$). A specificity-penalized prompt reduces the flip rate to 31%.

The expanded 1,000-item ConceptShift re-audit yields mean pairwise Cohen $\kappa=0.72$ on the binary fields `rater_1/2/3_correct` (no filtering), a high-confidence subset of 72% ($n=720$), majority accuracy 84.2%, and ambiguity rate 10.8%. Separately, applying the released surface-level classifier to *every* Self-Rewarding ConceptShift error on the test set attributes 52% to specificity bias against 5% for Base, in line with the 200-error human taxonomy (55% against 4%), and drops to 3% once CSD is applied.

H Certified-Only Training Sensitivity

OntoNeg-WN evaluation splits contain only certified denials. The released training pool mixes 284 certified rows with 1313 `kg_edge_absence` heuristics (Figure 7). Retraining Self-Rewarding on the certified-only subset over 3 seeds yields OntoNeg INA $44.27 \pm 0.76\%$ ($+17.87$ pp vs. Base) with ConceptShift AbsAcc $16.10 \pm 0.17\%$ (-43.40 pp; Table 26; Figure 4). Self-Critique, Entropy-Min, SFT-LoRA, Shuffled-DPO, and LoRA+KL under the same 3-seed protocol reproduce the sign reversal. Thus the local regression is *not* an artifact of missing-edge training labels; removing heuris-

tics enlarges both the target gain and the complementary collapse relative to the mixed-pool ledger in Table 2. The main ledger stays mixed-pool for cross-method comparability with previously released multi-seed controls.

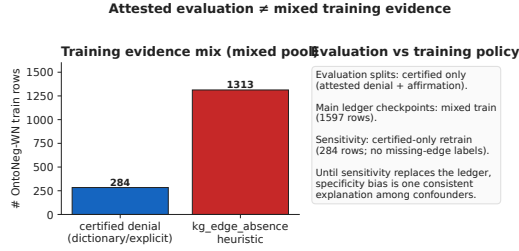


Figure 7: Training evidence mix versus attested-only evaluation. Main scores use certified evaluation splits; certified-only multi-seed retraining addresses the mixed-pool confounder.

I Ethics Statement

This statement covers human validation used in the paper. Benchmark *construction* itself is programmatic from WordNet and the ConceptNet 5.7 dump (Speer et al., 2017); FOLIO and ProofWriter are surface-style inspirations only and contribute no imported human-subject rows. Human judgments were collected only for quality audit (V1), an expanded ConceptShift re-audit, error taxonomy labeling, and direct-solver baselines (V4): 1,120 stratified audit items, 1,000 ConceptShift re-audit items (majority accuracy 84.2%, $\kappa=0.72$, ambiguity 10.8% on that re-audit alone), and $n=150$ items per task for human solvers (3 raters). Anonymized judgments are released in the audit package.

Annotator recruitment. Annotators were adult volunteers recruited through the authors’ institutional research channels in China (not a public crowdsourcing marketplace). The task was a short conceptual-reasoning / knowledge-item labeling exercise on public WordNet- and ConceptNet-derived items (gold vs. distractor / ambiguous judgments; binary correctness on the ConceptShift re-audit; taxonomy labels on sampled errors; and direct answers for the solver baseline). No personally identifying or sensitive demographic attributes were collected beyond voluntary contact information used solely for payment. Released files contain only anonymized rater ids.

Compensation. Annotators were paid in money (not course credit). The per-item rate was set so that the realized hourly equivalent met or exceeded the local prevailing wage for comparable short la-

Run (cert. train)	ON INA	CS AbsAcc
Base (reference)	26.40	59.50
Self-Rewarding mixed ledger	31.67	22.00
Self-Rewarding (3 seeds)	44.27 $\pm$ 0.76	16.10 $\pm$ 0.17
Δ vs Base	+17.87	-43.40
Self-Critique (3 seeds)	44.27 $\pm$ 1.21	14.03 $\pm$ 0.12
Δ vs Base	+17.87	-45.47
Entropy-Min (3 seeds)	52.33 $\pm$ 0.12	16.07 $\pm$ 0.47
Δ vs Base	+25.93	-43.43
SFT-LoRA (3 seeds)	43.80 $\pm$ 0.20	15.97 $\pm$ 0.12
Δ vs Base	+17.40	-43.53
Shuffled-DPO (3 seeds)	43.27 $\pm$ 0.61	15.73 $\pm$ 0.12
Δ vs Base	+16.87	-43.77
LoRA+KL (3 seeds)	43.80 $\pm$ 0.20	16.03 $\pm$ 0.06
Δ vs Base	+17.40	-43.47

Table 26: Certified-only OntoNeg training (284 rows; no kg_edge_absence). ON=OntoNeg, CS=ConceptShift. Means $\pm$ s.d. over completed seeds; all methods show OntoNeg gain with ConceptShift collapse. Main Table 2 remains the mixed-pool multi-method ledger.

beling work in the annotators’ region of residence in China; pilots were used to estimate minutes-per-item before finalizing the rate. The project record retains the per-item amount and the realized hourly equivalent used for that adequacy check. Participation could be stopped at any time without penalty to pay already earned for completed items.

Consent. Participation was voluntary. Before labeling, each annotator received written task instructions describing the item types, the judgment categories, the intended research use of anonymized labels in a scientific publication and public artifact release, and the absence of collection of sensitive personal attributes. Written informed consent was obtained for that research use. Annotators could skip items they found unclear and could withdraw; withdrawn incomplete work was not used.

IRB / ethics-review determination. Under our institution’s policy for this lightweight audit of public knowledge-base items (no intervention beyond short labeling; no sensitive personal data; adult volunteers), a full Institutional Review Board (IRB) / Research Ethics Board protocol was *not* required. The institutional determination / exemption wording for this protocol is retained in the project record. We did not treat the programmatic WordNet/ConceptNet construction pipeline as human-subjects research.

Use of AI Assistance. Contemporary OpenAI ChatGPT-class and Anthropic Claude-class assistants were used for language polishing of author-written drafts and for coding assistance on benchmark/experiment scripts. They were not used to

invent scientific claims, select which outcomes to report, invent numbers, or create unverified citations. All AI-assisted text and code were reviewed, tested, and validated by the authors, who take full responsibility for the manuscript and artifacts.

J Component Overview and Broader Impact

Total compute is approximately 48 GPU-hours on 2×RTX 4090D as recorded in the compute ledger (wall-clock hours multiplied by concurrent GPUs for training/eval jobs in that ledger; includes failed/debug runs where logged). We do not report an unverifiable carbon estimate. Benchmark construction uses public resources; human validation judgments are limited to the audits above. Risks are dominated by the underlying base model.

Component	Role	Primary evidence
DION-Bench Protocol	diagnostic substrate multi-axis evaluation	validity, robustness AuditPass, TOST
SR analysis	case study	objective bias
DION-Ref	low-drift reference	equivalence, drift

Table 27: Paper components and the evidence each contributes.

K Diagnostic Figures

This section collects the wide-format panels referenced from §4–§5: the locked-protocol signed-delta map and KL–regression scatter (Figure 8), hierarchical TOST equivalence intervals with the five-axis audit (Figure 9), and the specificity-bias composition, decoding response, complementary-axis mitigation, and nested-checkpoint scatters (Figures 10–11). Panel order follows the protocol—detect heterogeneity, test retention equivalence, then localize the complementary-axis failure mode.

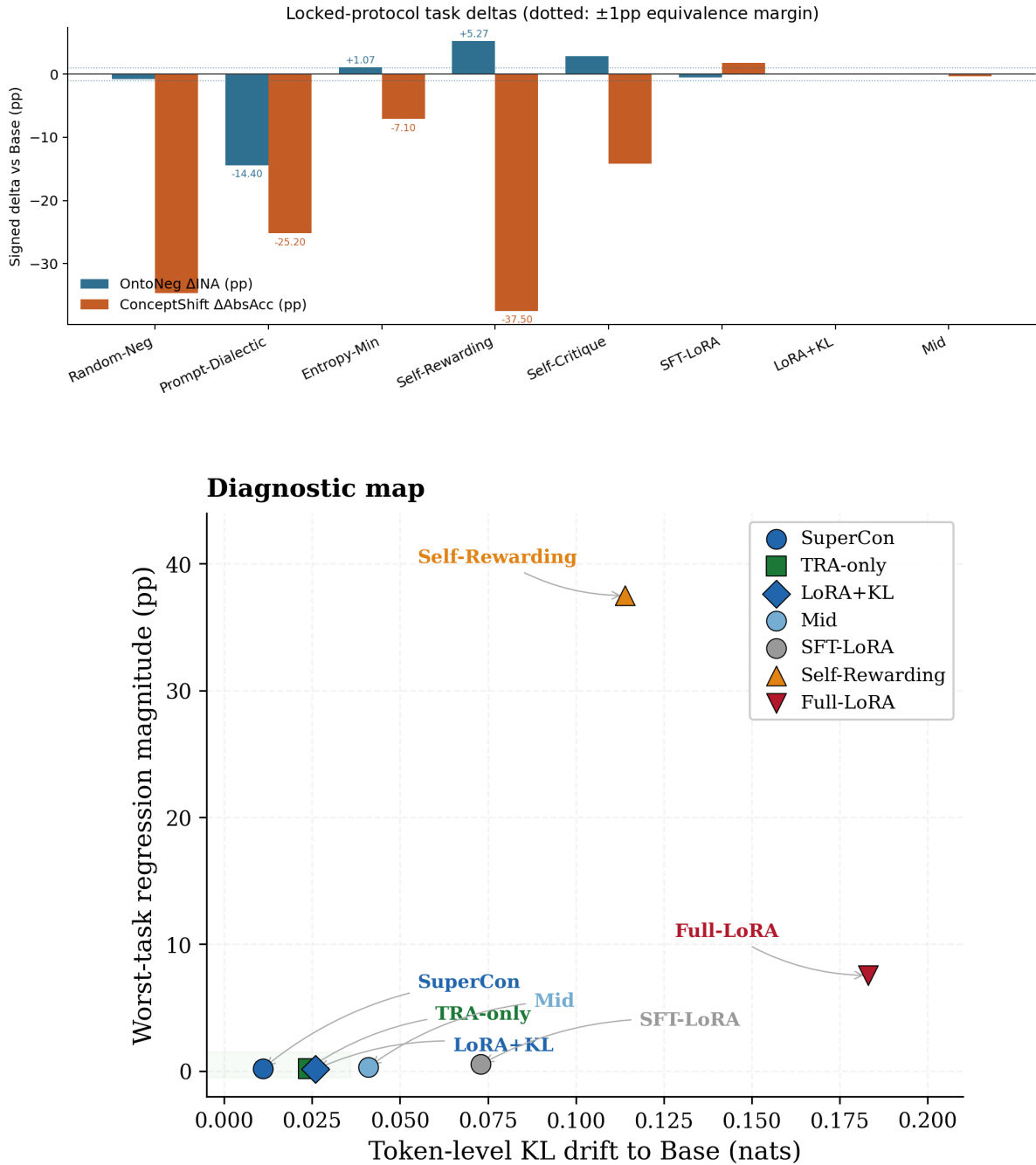
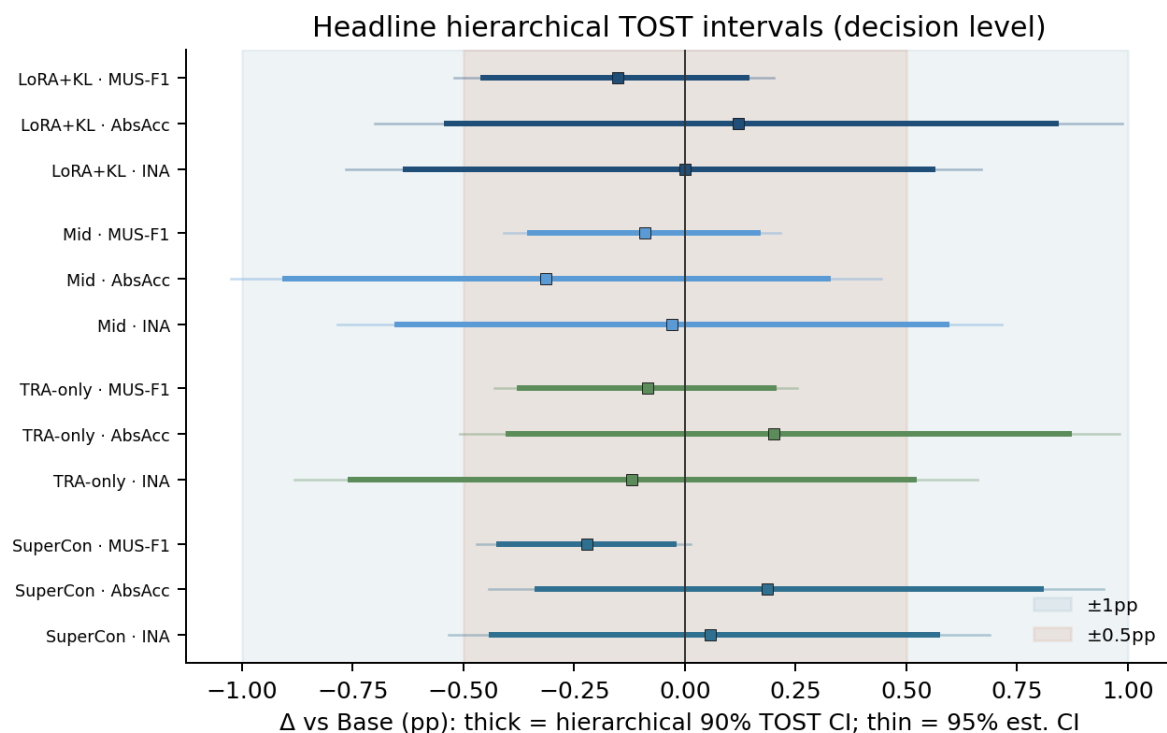


Figure 8: Primary diagnostic summary. *Top*: Signed task deltas vs Base (pp) from the released result ledger: Self-Rewarding shows $+5.27/-37.50$ heterogeneity; conservative DION-Ref and LoRA+KL stay inside the ± 1 pp band. *Bottom*: Scatter of token-level KL drift (x) against worst-task regression *magnitude* (y), defined as the largest absolute complementary-axis drop among OntoNeg OER, ConceptShift AbsAcc, and LogicRepair F1.



Equivalence audit (all five axes)

SuperCon	PASS	PASS	fail	fail
TRA-only	PASS	PASS	fail	fail
Mid	PASS	PASS	fail	fail
LoRA+KL	PASS	PASS	fail	fail
	±1 hier.	±1 clust.	±0.5 hier.	±0.5 item

Figure 9: Equivalence analysis aligned with the decision protocol. *Top*: Hierarchical (two-way) 90% TOST CI as thick segments (decision level), with thinner 95% estimation CI; square markers mark ± 1 pp pass. *Bottom*: All-five-axes audit at hierarchical/cluster/item levels. Item-level views are sensitivity only. Conservative references clear the pre-specified ± 1 pp operational-tolerance hierarchical/cluster gates and fail the tighter ± 0.5 pp sensitivity gates, matching Table 5 / the locked decision rule in §4.

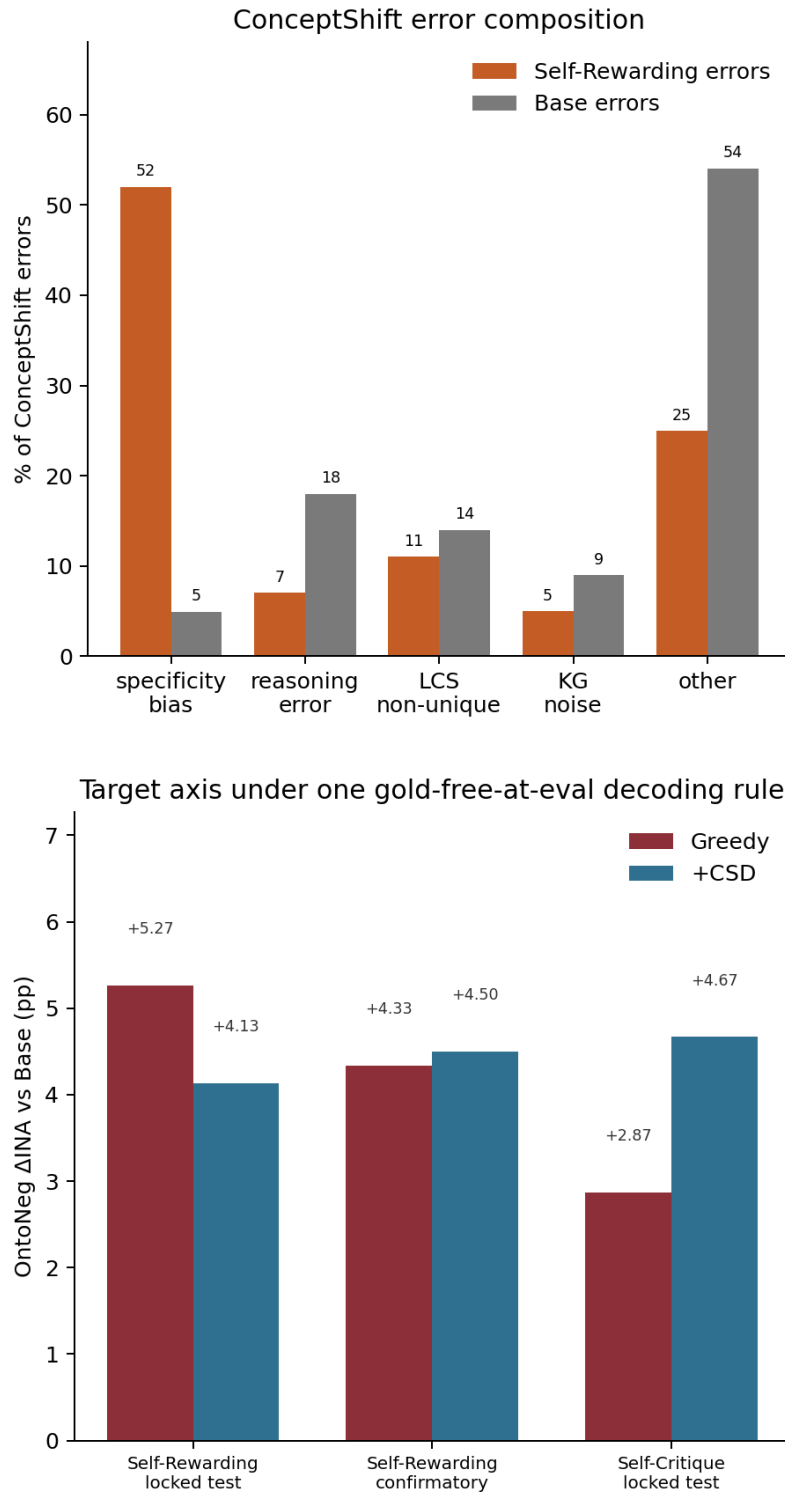


Figure 10: Specificity-bias diagnostics. *Top*: ConceptShift error composition—over-specific choices are common under Self-Rewarding (52%) and rare under Base (5%). *Bottom*: OntoNeg response of the same decoding rule on both splits. Complementary-axis recovery and nested-checkpoint scatters appear in Figure 11.

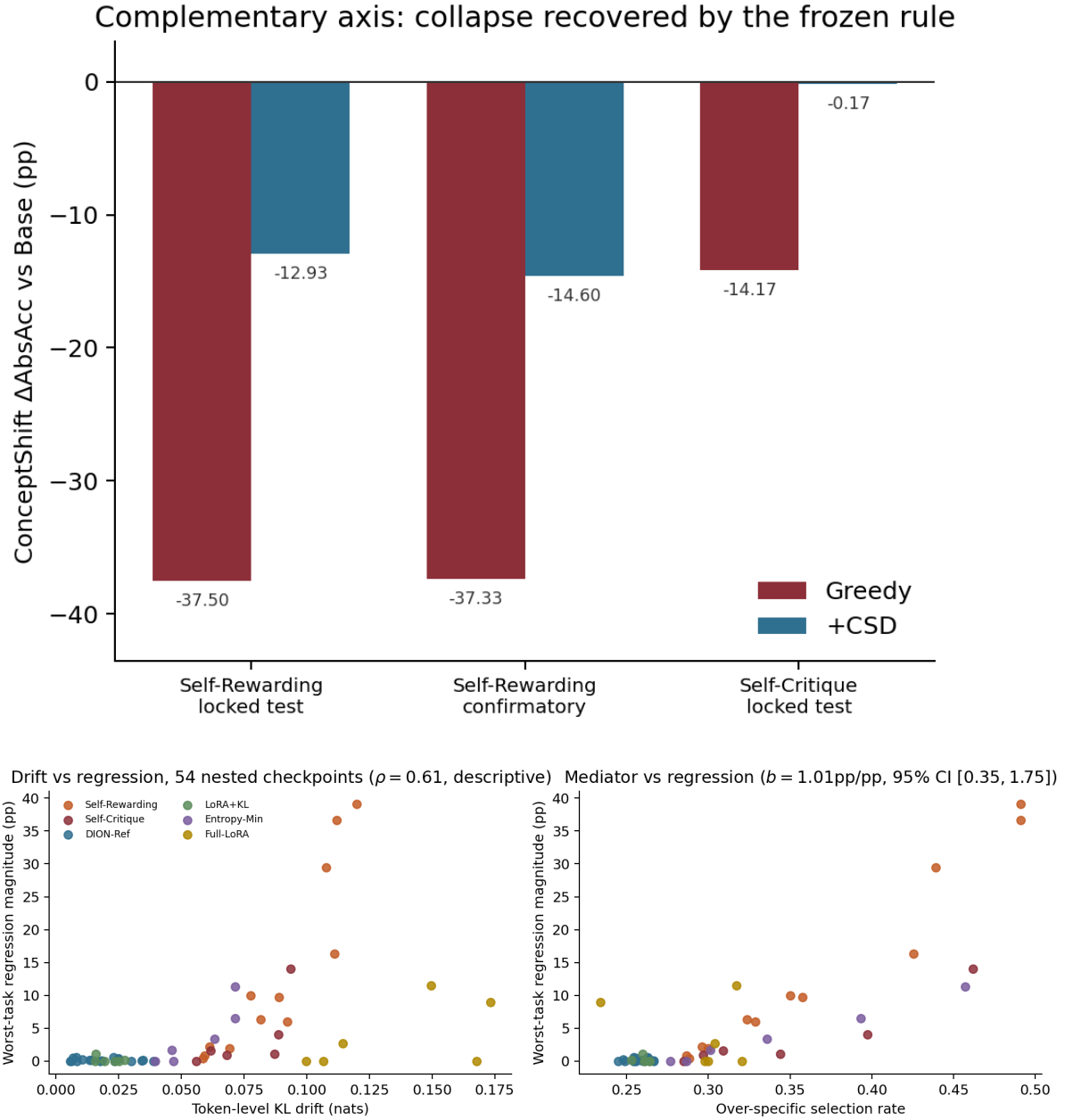


Figure 11: Specificity-bias mechanism (recovery and checkpoints). *Top*: complementary-axis regression the same rule reduces ($-37.50\text{pp} \rightarrow -12.93\text{pp}$ on the test set). *Bottom*: 54 nested, descriptive checkpoint observations (not independent samples) behind App. E.3: drift against worst-task regression, and over-specific selection rate against the same outcome. Checkpoint-level associations are trajectory diagnostics nested within runs; primary inference remains at the run-terminal ledger in §4.